

Final Master Thesis

Master's degree in Automatic Control and Robotics

Mathematical Modelling and Control applied to the Rocket Vertical Precision Landing Problem

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Author: José Alejandro Moya Blanco

Directors: Dr. Vicenç Puig Cayuela
Dr. Pablo Segovia Castillo

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Escola Tècnica Superior
d'Enginyeria Industrial de Barcelona



Abstract

The execution of autonomous vertical precision soft landing maneuvers has been the main way of achieving space rockets reusability during recent years. The solution to this problem has been developed for the last two decades and has been commercially and successfully implemented on orbital class rockets during the last decade. In order to tackle this problem from control systems perspective, this master thesis is dedicated to the study of some of the state of the art mathematical modelling and control engineering tools used to approach its solution. In order to do so, the thesis considers a theoretical single engine thrust vector controlled rocket based on the EmborockETH small-scale electric vertical takeoff vertical landing vehicle [1], and performs a full derivation of its three degrees of freedom and six degrees of freedom dynamic models with the objective of mathematically describing the three dimensional mechanics of the vehicle. With the objective of controlling the motion of the rocket, this thesis leverages previous work done in the guidance, navigation and control area to formulate a free-final-time optimal guidance and a nonlinear model predictive position control algorithms, which are intended to generate reference trajectories and thrust vector commands for the rocket to achieve soft landing maneuvers. Lastly and in order to assess the performance of these algorithms, both of them are implemented using Matlab, CasADi and IPOPT, and computationally simulated in diverse scenarios involving launch, landing and divert maneuvers.

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Acronyms

CoM	Center of Mass
DoF	Degrees of Freedom
G-FOLD	Guidance for Fuel Optimal Large Divert
GNC	Guidance, Navigation and Control
Hp	Prediction Horizon
IMU	Inertial Measurement Unit
MPC	Model Predictive Control
NMPC	Nonlinear Model Predictive Control
TVC	Thrust Vector Control
VTVL	Vertical Takeoff, Vertical Landing

Chapter 1

Introduction

The autonomous vertical precision landing of space rockets represents one of the most demanding and inspirational open problems to solve within the aerospace sector until the present day. The solution of this problem could lead to not only the reusability of space rockets for Earth launch operations, but also to groundbreaking improvements that would dramatically change the space exploration of the solar system as it is understood nowadays.

With the objective of modestly contributing to the solution of this challenging problem, this master thesis is dedicated to the study of some of the mathematical modelling and control engineering tools that enable the autonomous vertical precision landing of space rockets.



Figure 1.1: SpaceX Falcon Heavy Boosters Double Landing

1.1 Motivation

The idea of precisely landing space rockets emerges from two clear desires from the space flight industry sector: optimizing time demanding and very high cost processes that slow down the space exploration through spacecraft reusability and achieving the ability of reaching specific locations within nearby planets of the solar system in a highly precise manner.

From an engineering perspective, several disciplines of knowledge are involved in the design and building of a reusable space rocket: aerospace, mechanics, electronics, informatics, control systems among others. The latter is a multidisciplinary branch of engineering that is responsible for autonomously guiding and controlling the motion of the vehicle. Given the huge complexity of space rockets in general terms, the achievement of autonomous vertical precision landing maneuvers represent a big challenge for control engineering, and therefore, this discipline becomes really relevant when contributing to solving this problem.

Due to the fact that a large part of the contents of the Master's degree under which this thesis is being developed are related to Control Theory and Engineering, the autonomous vertical precision rocket landing problem is fully aligned with the learning objectives of the program.

1.2 Thesis Objectives

This work focuses on applying mathematical modelling and nonlinear optimal control techniques in order to achieve the generation and tracking of landing trajectories for a theoretical rocket based on the physical parameters of the EmborockETH VTVL vehicle [1]. The detailed milestones that are involved in the achievement of the global objective are:

- To derive a 3 degrees of freedom dynamic model of the rocket motion in 3D, a mathematically simple and lightweight description intended to be used in the guidance algorithm.
- To derive a 6 degrees of freedom dynamic model of the rocket motion in 3D, a more complex and accurate mathematical description intended to be used in the position control algorithm.
- To formulate a free-final-time optimal guidance algorithm intended to generate launch and landing trajectories for the rocket.
- To formulate a nonlinear model predictive position control algorithm intended to successfully track reference launch and landing trajectories.
- To implement both algorithms using Matlab and CasADI with the main objective of executing closed-loop rocket landing simulations in a 3D environment. In order to do so, in the first place, open-loop simulations are executed to validate the behavior of the dy-

dynamic models. Then, and in addition to landing simulations, launch maneuvers are also simulated with the objective of testing divert sequences, a integrated maneuver involving launch and landing, a typical test done in many relevant projects dealing with landing vehicles [2].

1.3 Thesis Structure

The content of this work is distributed as follows:

Chapter 2: History & State of the Art on Reusable Rockets

This chapter covers the history of the reusability of space launchers and describes its current state of the art. The most relevant academic publications and commercial projects related to the autonomous vertical precision rocket landing are reviewed.

Chapter 3: Mathematical Modelling of the Rocket

This chapter contains a detailed description of the theoretical rocket used for this work. In addition, the detailed derivation of the different vehicle's mathematical models is covered.

Chapter 4: Control Strategies

This chapter describes in detail the different optimal control strategies that are applied in order to achieve the generation, execution and simulation of rocket launch, landing and divert trajectories.

Chapter 5: Simulation Results

This chapter describes the implementation of both guidance and control algorithms, and also shows the results of different simulation cases along with their discussion.

Chapter 6: Effects on Economy, Society and Environment

This chapter briefly describes some possible impacts of the reusability of space launchers on society, economy and environment.

Chapter 7: Project Budget

This chapter summarizes the expenses related to this master thesis.

Chapter 8 : Concluding Remarks

This chapter shows the conclusions extracted during the realization of this work.

Chapter 2

History & State of the Art on Reusable Rockets

The concept of autonomous rockets has existed within the human knowledge for almost the last hundred years, but it was not until July 20, 1969 when the NASA Apollo 11 "Eagle" Lunar module spacecraft was half-autonomously landed in the surface of the Moon using a powered and controlled descent maneuver [3]. This milestone represents the first human approach to spacecraft powered landing, but at that time, rocket launchers were not yet believed to be capable of landing.

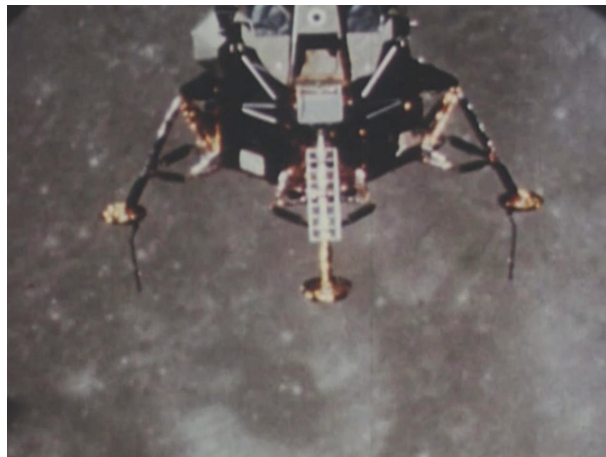


Figure 2.1: Apollo 11's lunar module, the Eagle, seen above the surface of the moon

The first human-made rocket launcher that achieved vertical landing maneuvers was the McDonnell Douglas DC-X [4], which took off and flew during 59 seconds on 18 August 1993 before finally landing using its attitude control and retro rocket thrusters. This milestone represents the first autonomous vertical rocket landing ever on mankind's history.



(a) Take-off



(b) Landing

Figure 2.2: McDonnell Douglas DC-X VTVL Rocket

Despite the cancellation of the previously mentioned project, it inspired the creation of many other new initiatives that pursued the development of groundbreaking reusability technologies that could be applied to the space flight industry. An example of this is represented by the Masten Space Systems Xombie XA-0.1B VTVL vehicle, a project that won the Northrop Grumman Lunar Lander Challenge (NG-LLC), funded by NASA, in 2009. The main objective of this project was to build a small-scale VTVL test vehicle that was controlled using a guidance, navigation and control (GNC) onboard system that actuated a thrust vectored liquid rocket engine. Many VTVL similar vehicles were built by Masten Space Systems between 2006 and 2016, as for instance Xoie, Xodiac or Xaero projects.



(a) Masten Xombie



(b) Masten Xodiac

Figure 2.3: Masten Space Systems VTVL test vehicles

During the last two decades and from a control engineering perspective, the autonomous vertical precision landing of space rockets has been approached using cutting-edge techniques that involve the use of mathematical optimization and model based control closed loops running on on-board GNC embedded systems. In this direction, the publication of "Lossless Convexification of Non Convex Control Bound and Pointing Constraints of the Soft Landing Optimal Control Problem" [5], by members of SpaceX and NASA Jet Propulsion Laboratory, represents a groundbreaking milestone that marked a turning point to the autonomous vertical precision landing of space rockets problem. This paper mainly described the use of a convexification process to transform nonconvex, hard to solve optimal control problems into equivalent convex problems, that were mathematically efficient and reliable to solve using convex optimization tools, which in general provide stronger theoretical guarantees in terms of stability, feasibility and convergence of the optimal control problem solution. Additionally, the use of convex optimization tools represents a potential improvement to real time on-board computations [5].

The previously mentioned research paper established the theoretical basis that led to the further empirical implementation of the G-FOLD GNC algorithm, presented in "G-FOLD: A Real-Time Implementable Fuel Optimal Large Divert Guidance Algorithm for Planetary Pinpoint Landing" [6]. In addition, and due to the fact that optimal control algorithms usually require high amounts of computational power to be solved, the recent technological improvements in on-board embedded computers has been a key factor for enabling the execution of this type of software in real-time applications. This powerful algorithm was empirically tested on 20 September 2013, as the result of a collaboration between NASA Jet Propulsion Laboratory and Masten Space Systems using the Xombie Lander vehicle [7], showing promising results.



Figure 2.4: On-board perspective during G-FOLD testing on Xombie Lander

Also during 2013, SpaceX commercial flight company started testing its Grasshopper VTVL prototype vehicle, that would later on end up transforming into the Falcon 9 Reusable launcher [8].



Figure 2.5: SpaceX Grasshopper Testing

After several years of active research and thanks to all the knowledge collected during the previously mentioned projects, reusable launchers operated by commercial space flight companies are now a reality. The main companies that currently offer space launch services for payload or human crew missions are SpaceX, through Falcon 9 and Falcon Heavy launchers, and Blue Origin, with its New Shepard rocket.



(a) Falcon 9 Reusable Booster being towed after landing



(b) New Shepard Reusable Booster vertically landing

Figure 2.6: Commercial Reusable Rockets - Falcon 9 and New Shepard

Furthermore, several research projects are being actively developed in the present in relation to the reusability of different sized rocket launchers, with the objective of carrying out a wide range of space flight missions. Some of the most relevant ones are: SpaceX Starship, Blue Origin New Glenn, DLR-CNES-JAXA Callisto [9], Rocket Lab Electron, LandSpace Zhuque-3, Mast Space Systems Xogdor, among others.



(a) SpaceX Starship (121 m height) launch



(b) LandSpace Zhuque-3 Divert test

Figure 2.7: SpaceX Starship and LandSpace Zhuque-3

Finally, and from the control academic field point of view, the autonomous rocket vertical precision landing is actively being researched as there are still several open problems whose solution involve the improvement of nonlinear control, optimal control, robust control and mathematical modelling methods.

In order to do so, some academic projects pursue the design and implementation of cheap small-scale electric propelled rocket test beds that can be used to study new guidance, navigation and control algorithms in a quick and cost effective manner.



(a) EmborockETH from ETH Zurich [1]

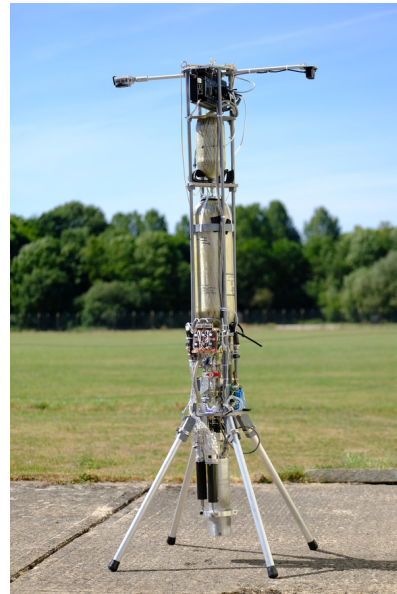


(b) Umouche from EPFL GSP [10]

Moreover, some other state of the art projects pursue the design and implementation of small-scale pure rocket test beds, which actually use liquid rocket engines:



(a) Collibri from EPFL GSP [10]



(b) Gyroc from Airborne Engineering LTD [11]

Chapter 3

Mathematical Modelling of the Rocket

Dynamic models are one of the key mathematical tools used to study the laws of motion of complex physical objects. These models are usually a simplified representation of the underlying physics of the system and describe how relevant properties of the system change over time by using continuous-time differential equations.

With the objective of characterizing the dynamic properties of a rocket as a vehicle, during this section of the thesis, the dynamics of a single engine thrust vector controlled rocket are studied by using the tools provided by the mathematical theory of rigid bodies.



Figure 3.1: SpaceX Grasshopper & Falcon using single engine TVC configuration

3.1 Vehicle General Description

The main case study used in this work consists of a theoretical single engine thrust vector controlled rocket that can freely fly within the 3D space.

This type of vehicles usually present a cylindrical shape and use one or several thrusters located at its bottom to produce movement by reaction, as stated by Newton's third law of motion.

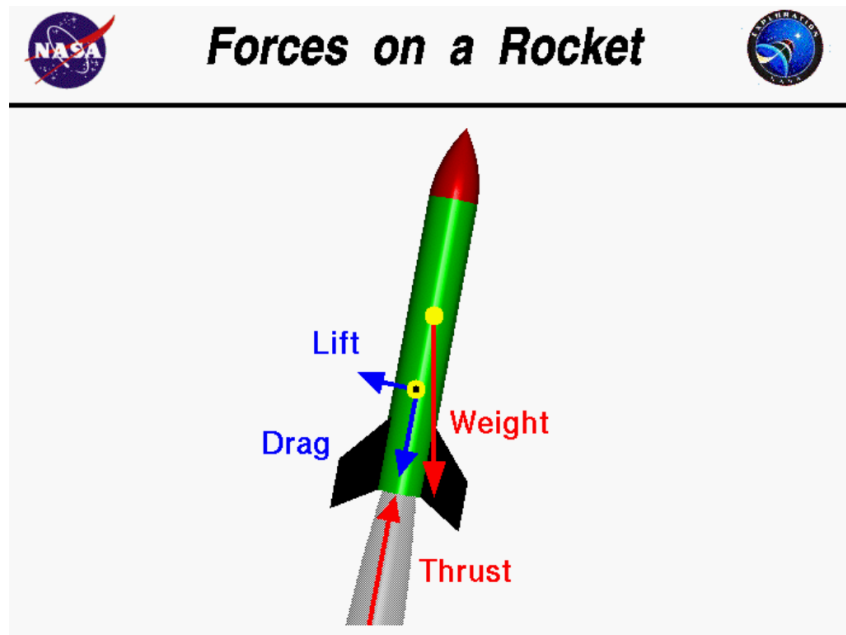


Figure 3.2: Simplified Forces Diagram of a Rocket by NASA [12]

In order to control the flight direction of the vehicle, several strategies have been proposed, being thrust vector control (TVC) one of the most widely used. Due to the fact that the center of mass of the rocket is usually located within its longitudinal axis at a certain distance of the thruster, this strategy consists in the capability of varying the pointing direction of the produced thrust with the objective of not only generating translational forces along its longitudinal axis, but also rotational torques around its center of mass, and therefore, allowing the control of its flight direction (position + orientation).

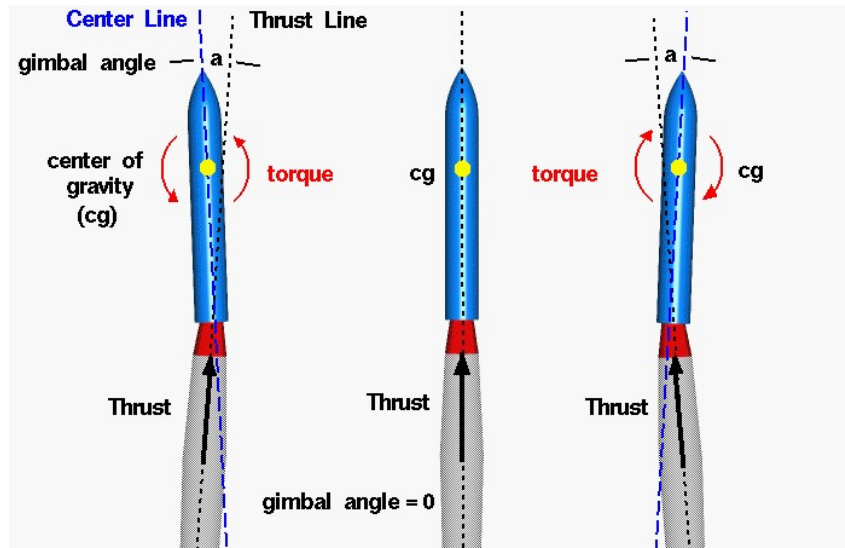


Figure 3.3: Thrust Vector Control Diagram of a rocket by NASA [12]

It is important to remark that, in order to simplify the vehicle modelling, no other sources of control apart from the main engine thrust vectoring are considered in this work. In addition, aerodynamic forces (lift and drag) are not taken into account in the mathematical model and therefore the vehicle is assumed not to have any fins.

Although the content of this master thesis is developed in a theoretical manner and with the objective of simulating the dynamics of a physically built rocket, the vehicle used in this work is based on the EmborockETH [1]. This project was developed at ETH Zurich Institute's for Dynamic Systems and Control, and aims to build a small scale VTOL vehicle that mimics the physical configuration and dynamics of the first stage of a reusable rocket launcher controlled by a gimbaled electric propeller. Its physical structure is as follows:

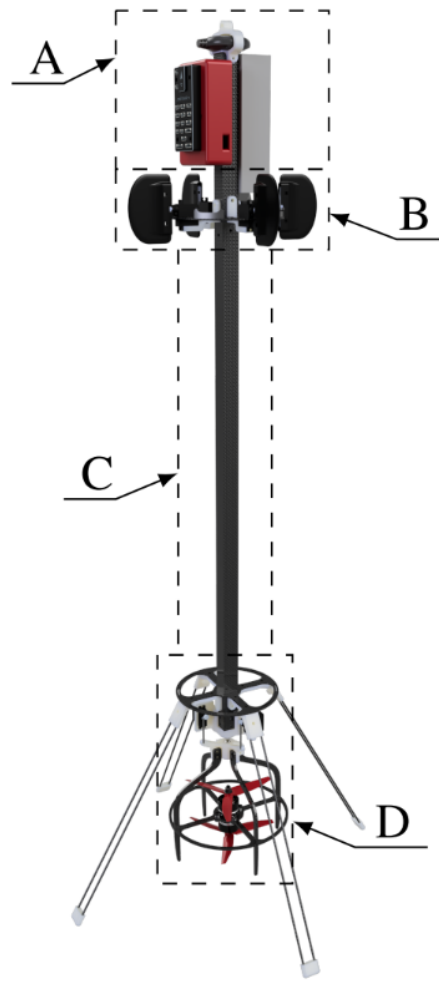


Figure 3.4: EmborockETH vehicle by ETH Zurich [1] - A: Electronics & Battery, B: Aerodynamic Fins, C: Central Spine, D: Gimbal Mechanism and Electric Propeller

3.2 Coordinate System

Due to the fact that the rocket is intended to fly during short periods of time and within small distances, a logical assumption is to consider a non-rotating and flat earth environment as in other works [13].

In order to represent the position and orientation of the rocket within the 3D Euclid space, two reference frames are commonly used:

- An inertial reference frame, which is considered to be the world reference frame.
- A non-inertial and rotating reference frame, which is considered to be the vehicle reference frame, whose origin is fixed to its center of mass.

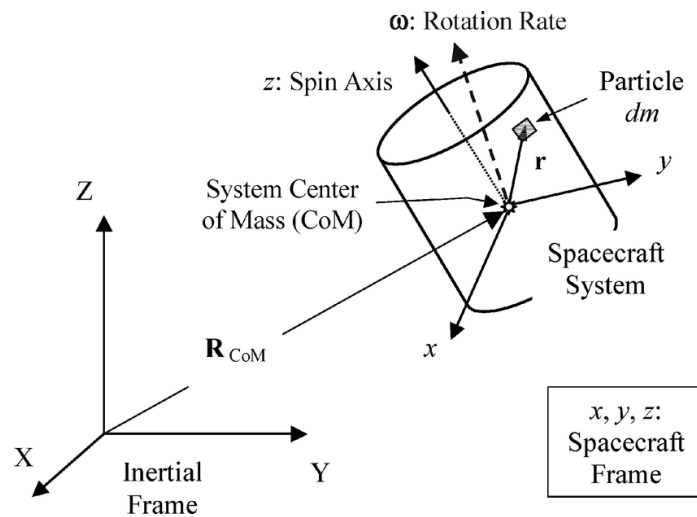


Figure 3.5: Example of Inertial/Vehicle reference frames representation in aerospace [14]

In this work, the orientation of the vehicle frame with respect to the world frame at rest conditions is such that the longitudinal axis of the rocket is coincident with the x axis and pointing up, being the z axis aligned with the forward movement and the y axis with the right movement:

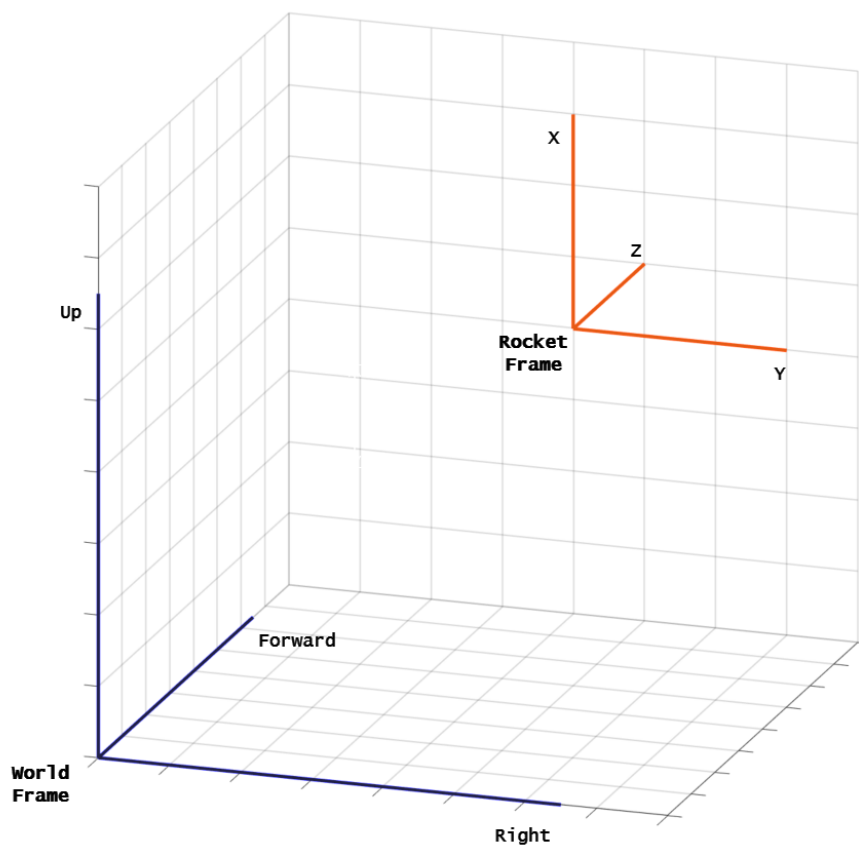


Figure 3.6: World and Rocket reference frames

When no explicitly indicated, SI units are considered when describing physical properties of the vehicle.

3.3 Three Degrees of Freedom Vehicle model

In this work, the vehicle is considered a rigid body and therefore, its motion is determined by the translational and rotational forces acting on it, which are caused by the vectored thrust and gravity.

Initially and as described in previous works [1] [13], a simple mathematical description of the vehicle using a point mass model is derived. In this model, only the translational motion of the vehicle is considered, and therefore its rotational motion is neglected. Due to this fact, this model is considered a three degrees of freedom (3 DoF) model since it only contemplates the position of the vehicle and not its orientation.

It is important to remark that, even if this model does not represent the full dynamics of the vehicle and misses a relevant part of the real motion of the body, it provides a mathematically lightweight formulation that will be really useful for future chapters of this thesis.

When considering the vehicle as a point mass body, its translational dynamics can be adequately described using Newton's second law of motion, which states the following:

$$F = m \cdot a = m \cdot \dot{v} \quad (3.1)$$

By applying this equation on each orthogonal principal axis, the following state space model can be formulated:

$$\begin{aligned} \dot{p}_x &= v_x \\ \dot{p}_y &= v_y \\ \dot{p}_z &= v_z \\ \dot{v}_x &= \frac{T_x}{m} + g \\ \dot{v}_y &= \frac{T_y}{m} \\ \dot{v}_z &= \frac{T_z}{m} \end{aligned} \quad (3.2)$$

Notation remark: The explicit dependence of state and input variables on time is omitted for convenience

The state vector is $x = [p_x, p_y, p_z, v_x, v_y, v_z]$ where p_x, p_y, p_z represent the position of the vehicle in the 3D space, and v_x, v_y, v_z represent its velocity vector.

The input vector is $T = [T_x, T_y, T_z]$ which is the thrust 3D vector produced by the main engine.

Additionally, the following vectors are defined: $p = [p_x, p_y, p_z]$ and $v = [v_x, v_y, v_z]$.

The resulting mathematical model is linear.

3.4 Six degrees of freedom vehicle model

As mentioned in the Section 3.3, the 3 DoF model partially models the dynamics of the vehicle, taking into account its translational dynamics but neglecting its rotational dynamics. This model may be precise enough for simple use cases but, in general terms, lacks accurateness in terms of rigorously describing the full physics behind the motion of the vehicle. Due to this fact, a more elaborated mathematical model is required when describing its translational and rotational dynamics in an aggregated manner, as done in [1] [13] [15] [16].

As a result of adding the vehicle's rotational motion to the mathematical model, new degrees of freedom emerge, as the pointing direction of the rocket can now change. Due to this fact, the six degrees of freedom (6 DoF) model aims to describe the position and orientation (also known as "attitude" in aerospace engineering) of the vehicle.

3.4.1 Rotational Dynamics

The rotational dynamics of the vehicle describe how it can rotate in the 3D space and can be modelled using the Euler's equations described in the rigid body dynamics classical theory, which state the following:

$$J \cdot \dot{\omega} + \omega \times (J \cdot \omega) = \tau \quad (3.3)$$

When dealing with the previous equation, the inertia matrix J determines the body's resistance to rotation, which in this work takes the following form:

$$J = \begin{bmatrix} J_{xx} & 0 & 0 \\ 0 & J_{yy} & 0 \\ 0 & 0 & J_{zz} \end{bmatrix}$$

In order to particularize the previous equation, the torque is expressed in terms of the thrust vector:

$$J \cdot \dot{\omega} + \omega \times (J \cdot \omega) = T \times r_{CoM} \quad (3.4)$$

In this case, the previous equation is intended to describe the 3D rotational dynamics of a rigid body and therefore, it is useful to formulate it for the orthogonal principal axes of inertia, becoming the following:

$$\begin{aligned} J_{xx} \cdot \dot{\omega}_x + (J_{zz} - J_{yy}) \cdot \omega_y \cdot \omega_z &= T_y \cdot 0 - T_z \cdot 0 \\ J_{yy} \cdot \dot{\omega}_y + (J_{xx} - J_{zz}) \cdot \omega_z \cdot \omega_x &= T_z \cdot r_{CoM} - T_x \cdot 0 \\ J_{zz} \cdot \dot{\omega}_z + (J_{yy} - J_{xx}) \cdot \omega_x \cdot \omega_y &= T_x \cdot 0 - T_y \cdot r_{CoM} \end{aligned} \quad (3.5)$$

From the previous equations, the following partial state space formulation can be extracted:

$$\begin{aligned} \dot{\omega}_x &= \frac{1}{J_{xx}} \cdot (T_y \cdot 0 - T_z \cdot 0) - (J_{zz} - J_{yy}) \cdot \omega_y \cdot \omega_z \\ \dot{\omega}_y &= \frac{1}{J_{yy}} \cdot (T_z \cdot r_{CoM} - T_x \cdot 0) - (J_{xx} - J_{zz}) \cdot \omega_z \cdot \omega_x \\ \dot{\omega}_z &= \frac{1}{J_{zz}} \cdot T_x \cdot 0 - T_y \cdot r_{CoM} - (J_{yy} - J_{xx}) \cdot \omega_x \cdot \omega_y \end{aligned} \quad (3.6)$$

An important remark is that r_{CoM} represents the position offset between the vehicle's center of mass and the thrust vector application point. It is considered to be expressed as a position vector in 3.4 and an scalar value in 3.5, 3.6.

3.4.2 Attitude Dynamics

Vehicle attitude describes its orientation in the 3D space, that varies as a function of the rotation torques that affect the center of mass of the body. Due to this fact, the attitude dynamics are a direct consequence of the rotational dynamics.

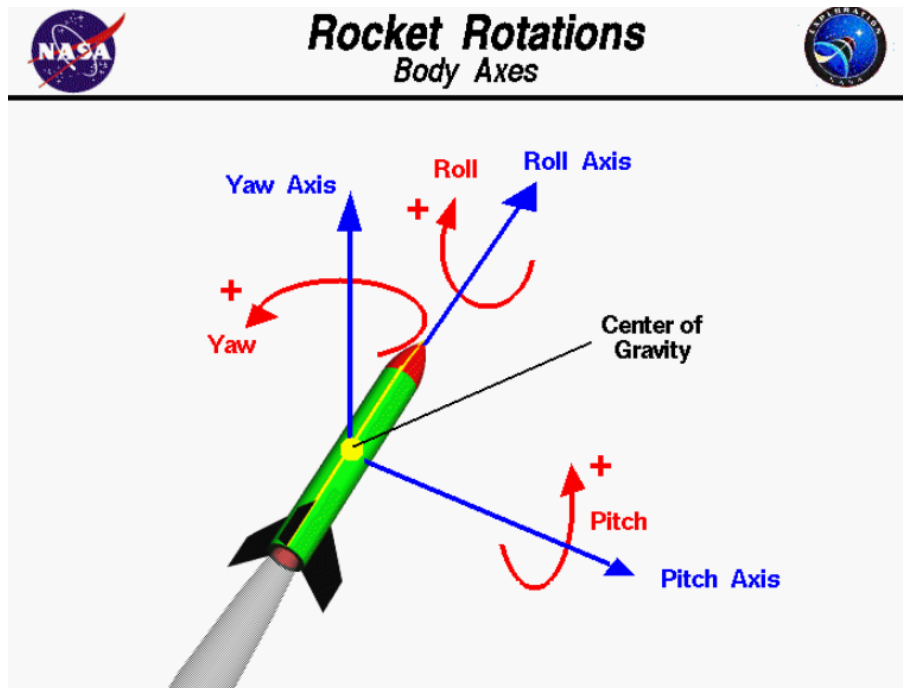


Figure 3.7: Rocket Rotations Diagram by NASA [12]

The two main ways of mathematically expressing the attitude of a rigid body are the following:

- Euler angles, which determine the orientation of a rigid body in the 3D space with respect to a fixed frame in terms of the inclination angle of each orthogonal principal axis, described by three values that can be ordered in several manners and that indicate roll, yaw, pitch sequence rotations.
- Quaternions, a 4D extension of regular complex numbers that provide a convenient notation for describing 3D space rotations when treating them as versors [17]. Using this particular type of quaternions, the orientation of a rigid body can be described in the 3D space using four numeric values.

In general terms, quaternions are considered to be more complex mathematical objects when compared with Euler angles, and due to this fact, their properties provide several advantages when describing the attitude of a rigid body. Some of the most relevant are listed below:

- Quaternions avoid singularities when describing spatial orientation, and therefore, fix the gimbal lock issue [18].
- Thanks to their mathematical formulation, quaternions avoid the need to compute trigonometric functions, and also provide a more computationally efficient set of mathematical operations.

- Nowadays, quaternions are considered the state of the art method for representing the attitude of spacecrafts, aircrafts, robots and other similar applications, and therefore, using this method eases the integration with other existing tools.

Due to the previous facts and in order to grant robust and efficient attitude representations, the spatial orientation of the vehicle considered in this work will be described using unit quaternions.

Given the fact that the attitude of the vehicle may change over time due to its rotational dynamics, the quaternion theory literature [19] [20] states that the first order approximation for the attitude dynamics is formulated in the following manner:

$$\dot{q} = \frac{1}{2} \cdot \Omega(\omega) \cdot q, \quad (3.7)$$

where the Ω operator [19] describes how the rotational dynamics affect the attitude dynamics and is defined as follows:

$$\Omega(\omega) \triangleq [\omega]_R = \begin{bmatrix} 0 & -\omega^\top \\ \omega & -[\omega]_\times \end{bmatrix} = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix}$$

By developing the previous quaternion dynamics expression for each of its components, the following partial state space representations is deduced:

$$\begin{aligned} \dot{q}_w &= \frac{1}{2} \cdot (-\omega_x \cdot q_x - \omega_y \cdot q_y - \omega_z \cdot q_z) \\ \dot{q}_x &= \frac{1}{2} \cdot (\omega_x \cdot q_w + \omega_z \cdot q_y - \omega_y \cdot q_z) \\ \dot{q}_y &= \frac{1}{2} \cdot (\omega_y \cdot q_w - \omega_z \cdot q_x + \omega_x \cdot q_z) \\ \dot{q}_z &= \frac{1}{2} \cdot (\omega_z \cdot q_w + \omega_y \cdot q_x - \omega_x \cdot q_y) \end{aligned} \quad (3.8)$$

Two key assumptions are taken when dealing with the attitude dynamics:

- The norm of the quaternion should be kept unitary over time in order to maintain its consistency for describing vehicle attitude.
- As the quaternion dynamics are expressed using a first order Taylor approximation, the

variations in angular velocity affecting the Ω operator should be as small as possible in order to avoid potential numerical errors when integrating them.

- The quaternion components convention used in this work is the Hamilton ordering, as described in [19].

3.4.3 Translational Dynamics

Given the fact that in the six degrees of freedom model the vehicle can now rotate in the 3D space, the translational dynamics are now affected by the attitude dynamics, as the point of application of the thrust rotates along with the vehicle.

As mentioned in Section 3.4.2, the attitude of the vehicle is determined by the four components of the attitude quaternion, which can be transformed to a equivalent rotation matrix that can be used to adequately rotate the vehicle frame.

Given the attitude quaternion, the equivalent rotation matrix is formulated as follows:

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_z q_w) & 2(q_x q_z + q_y q_w) \\ 2(q_x q_y + q_z q_w) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_x q_w) \\ 2(q_x q_z - q_y q_w) & 2(q_y q_z + q_x q_w) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix}$$

where q_w, q_x, q_y, q_z are the components of the attitude quaternion determined by the attitude dynamics.

In order to adapt the 3-DoF translational dynamics to take into account the attitude degrees of freedom, the thrust vector needs to be rotated in function of the vehicles attitude, producing the following formulation:

$$\begin{aligned} \dot{\mathbf{p}} &= \mathbf{v} \\ \dot{\mathbf{v}} &= \mathbf{R}(\mathbf{q}) \cdot \frac{T}{m} + \mathbf{g} \end{aligned} \tag{3.9}$$

By developing the previous equations for the orthogonal principal axes, the following partial state space formulation is obtained:

$$\begin{aligned}
\dot{p}_x &= v_x \\
\dot{p}_y &= v_y \\
\dot{p}_z &= v_z \\
v_x &= \frac{1}{m} \cdot \left(\left(1 - 2(q_y^2 + q_z^2) \right) \cdot T_x + \left(2(q_x q_y - q_z q_w) \right) \cdot T_y + \left(2(q_x q_z + q_y q_w) \right) \cdot T_z - m \cdot g \right) \\
v_y &= \frac{1}{m} \cdot \left(\left(2(q_x q_y + q_z q_w) \right) \cdot T_x + \left(1 - 2(q_x^2 + q_z^2) \right) \cdot T_y + \left(2(q_y q_z - q_x q_w) \right) \cdot T_z \right) \\
v_z &= \frac{1}{m} \cdot \left(\left(2(q_x q_z - q_y q_w) \right) \cdot T_x + \left(2(q_y q_z + q_x q_w) \right) \cdot T_y + \left(1 - 2(q_x^2 + q_y^2) \right) \cdot T_z \right)
\end{aligned} \tag{3.10}$$

3.4.4 Complete State Space Formulation

By combining the rotational, attitude and translational partial state space dynamics, the complete 6-DoF vehicle dynamics expressed as a nonlinear state space model can be formulated:

$$\left\{ \begin{aligned}
\dot{p}_x &= v_x \\
\dot{p}_y &= v_y \\
\dot{p}_z &= v_z \\
v_x &= \frac{1}{m} \cdot \left(\left(1 - 2(q_y^2 + q_z^2) \right) \cdot T_x + \left(2(q_x q_y - q_z q_w) \right) \cdot T_y + \left(2(q_x q_z + q_y q_w) \right) \cdot T_z + m \cdot g \right) \\
v_y &= \frac{1}{m} \cdot \left(\left(2(q_x q_y + q_z q_w) \right) \cdot T_x + \left(1 - 2(q_x^2 + q_z^2) \right) \cdot T_y + \left(2(q_y q_z - q_x q_w) \right) \cdot T_z \right) \\
v_z &= \frac{1}{m} \cdot \left(\left(2(q_x q_z - q_y q_w) \right) \cdot T_x + \left(2(q_y q_z + q_x q_w) \right) \cdot T_y + \left(1 - 2(q_x^2 + q_y^2) \right) \cdot T_z \right) \\
\dot{q}_w &= \frac{1}{2} \cdot (-\omega_x \cdot q_x - \omega_y \cdot q_y - \omega_z \cdot q_z) \\
\dot{q}_x &= \frac{1}{2} \cdot (\omega_x \cdot q_w + \omega_z \cdot q_y - \omega_y \cdot q_z) \\
\dot{q}_y &= \frac{1}{2} \cdot (\omega_y \cdot q_w - \omega_z \cdot q_x + \omega_x \cdot q_z) \\
\dot{q}_z &= \frac{1}{2} \cdot (\omega_z \cdot q_w + \omega_y \cdot q_x - \omega_x \cdot q_y) \\
\dot{\omega}_x &= \frac{1}{J_{xx}} \cdot (T_y \cdot 0 - T_z \cdot 0) - (J_{zz} - J_{yy}) \cdot \omega_y \cdot \omega_z \\
\dot{\omega}_y &= \frac{1}{J_{yy}} \cdot (T_z \cdot r_{CoM} - T_x \cdot 0) - (J_{xx} - J_{zz}) \cdot \omega_z \cdot \omega_x \\
\dot{\omega}_z &= \frac{1}{J_{zz}} \cdot (T_x \cdot 0 - T_y \cdot r_{CoM}) - (J_{yy} - J_{xx}) \cdot \omega_x \cdot \omega_y
\end{aligned} \right. \tag{3.11}$$

Notation remark: The explicit dependence of state and input variables on time is omitted for convenience

The state vector is $x = [p_x, p_y, p_z, v_x, v_y, v_z, q_w, q_x, q_y, q_z, \omega_x, \omega_y, \omega_z]$ where p_x, p_y, p_z represent the linear position of the vehicle in the 3D space, v_x, v_y, v_z represent its linear velocity vector, q_w, q_x, q_y, q_z represent the vehicles attitude and $\omega_x, \omega_y, \omega_z$ represent its angular velocity vector.

The input vector is $T = [T_x, T_y, T_z]$ which is the thrust 3D vector produced by the main engine.

Additionally, the following vectors are defined: $p = [p_x, p_y, p_z]$, $v = [v_x, v_y, v_z]$, $q = [q_w, q_x, q_y, q_z]$ and $\omega = [\omega_x, \omega_y, \omega_z]$.

The resulting mathematical model is strongly nonlinear and presents bilinear terms involving states and inputs. This can be a problem when trying to use control methods other than non-linear ones, such as linear parameter-varying (LPV) control.

Finally, the full state of the rocket vehicle is assumed to be observable during the theoretical developments of this work.

Note: A more compact formulation of 3.11 can be found in 4.3.3.

Chapter 4

Control Strategies

As stated in Section 1.2, once the dynamics of the rocket vehicle are precisely described using mathematical models, the idea is to use such models to control its motion and attitude with the objective of performing launch and landing maneuvers. An approach to solving this problem consists in using tools from Control Theory.

Given the fact that precise mathematical models of the vehicle are available and leveraging state of the art work done in the GNC domain, model-based control strategies will be used along this section of the thesis.

4.1 Control Problem Description

From an engineering perspective, the situation subject of this work can be understood as a control problem in which the dynamics of the vehicle should be adequately controlled using actuators in order to achieve a desired behavior. Specifically, the rocket guidance and control problem consists in controlling its motion so that it moves from initial point to a target point, where a target position, velocity and orientation needs to be achieved. A particular case of this problem is the rocket soft vertical precision landing problem, which consists in controlling the motion of the rocket so that it hits zero altitude with zero velocity and vertical pointing up orientation.

With the objective of controlling the motion and attitude of the rocket in a desired manner, it is important to remember that the considered vehicle in this work is a small-scale rocket-like vehicle which uses a single thrust vector controlled engine configuration. Due to this fact and as described in Section 3.1, all the control capabilities of the vehicle depend on the use of the thrust vector to produce the proper variations in the dynamic model of the rocket.

In general terms, and when dealing with rocket control problems, two underlying problems are posed:

- Vehicle's guidance, which aims to find feasible trajectories for the vehicle to follow so that it performs the desired maneuvers. In addition, the calculated trajectories are usually intended to be optimal with respect to some criteria, as for example: fuel consumption, flight distance, flight time, etc; and also are usually constrained by additional mission requirements.
- Vehicle's control, which, once a given trajectory is calculated, aims to govern the vehicle's dynamics through the existent control inputs in order to track the planned reference trajectory with minimal error.

These two problems are often solved using a multilayer control scheme, where the guidance is solved at the highest layers using lower frequency processes. Then, the generated guidance results are used by the lower reference tracking layers, which attempt to control the behavior of the vehicle using higher frequency processes. Both problems are usually solved in an online manner, and therefore are classified as critical real-time control systems.

In this section of the thesis, a solution to both problems is theoretically proposed as well as implemented using code.

4.2 Free-Final-Time Optimal Guidance

As a first step in the control strategies for this work, an optimal guidance with free-final-time trajectory generator is proposed. The main objective of this subsystem is to generate a valid trajectory that connects two points in the 3D space by taking into account a set of initial and final conditions for the vehicle to comply with, and by also verifying some extra optimality criteria and flight constraints.

The key feature of the guidance subsystem subject of this work consists in its capability to generate optimal trajectories, which are achieved by formulating the guidance problem as a model-based optimal control problem, which is solved by using tools from mathematical optimization theory.

Specifically, the guidance system presented here is intended to compute time-optimal and energy-optimal trajectories for the vehicle to follow by using the 3-DoF vehicle model derived in Section 3.3. The use of this simplified model instead of the 6-DoF vehicle model aims to minimize the computational power required to solve the model-based optimal control problem by keeping the essential translational dynamics of the vehicle, which represent a good enough approximation as stated in previous works related with this field of knowledge [1] [13] [16].

It is important to remark that the free-final-time feature within the optimal guidance subsystem represents a state of the art result to approach this problem, in comparison with older works where only the fuel-optimality or energy-optimality criteria were contemplated while using a

fixed-final-time formulation [15], as it enables the introduction of the time-optimality criterion into the trajectory generation process. From an intuitive point of view, this improvement provides the guidance subsystem with the capability of calculating trajectories that minimize the vehicle's flight time along the desired maneuver.

4.2.1 Time-scaled 3-DoF Dynamic Model

For any generated trajectory, its elapsed time is defined by the time interval $[t_0, t_f]$, where t_0 represents the initial time instant of the trajectory, usually assumed to take the value $t_0 = 0$ s, and t_f represents the final time instant. In order to achieve time-optimal trajectories with minimal flight time, the value of t_f should be as small as possible.

Given the fact that the guidance subsystem generates trajectories by solving a model based optimal control problem, optimality criteria are represented as optimization variables within this problem and should explicitly appear within the vehicle's dynamics. This means that in order to minimize the flight time, the variable t_f should be added to the 3-DoF vehicle's model formulation.

For any calculated trajectory, the differential equations described in Section 3.2 provide the evolution of the vehicle's dynamics in the range $[0, t_f]$, but as it can be seen, the variable t_f does not explicitly appear within the equations, and therefore cannot be treated as an optimization variable in the optimal guidance problem. As stated in [1] [21] [22], a smart reformulation of the problem consists in changing the time scale of the initial differential equation by normalizing its original time range $[0, t_f]$, where the value of t_f is free, to a fixed time range $[0, 1]$, and therefore forcing the introduction of the variable t_f as an explicit multiplicative factor of the whole differential equation.

Given the following general state space formulated differential equation:

$$\dot{x} = f(x, u) \equiv \frac{dx}{dt} = f(x, u) \text{ where } t \in [0, t_f], \quad (4.1)$$

the differential equation can be time-scaled as previously mentioned by introducing the variable change

$$dt = d\theta \cdot t_f, \quad (4.2)$$

which yields

$$\frac{dx}{d\theta} = f(x, u) \cdot t_f, \quad (4.3)$$

where now $\theta \in [0, 1]$. The time variable t is then replaced by a time-scaled variable θ , which allows the required free-final-time reformulation.

By applying the previous idea, the following time scaled reformulation for the 3-DoF is obtained:

$$\begin{aligned}
 \dot{p}_x &= v_x \cdot t_f \\
 \dot{p}_y &= v_y \cdot t_f \\
 \dot{p}_z &= v_z \cdot t_f \\
 \dot{v}_x &= \left(\frac{T_x}{m} + g \right) \cdot t_f \\
 \dot{v}_y &= \left(\frac{T_y}{m} \right) \cdot t_f \\
 \dot{v}_z &= \left(\frac{T_z}{m} \right) \cdot t_f
 \end{aligned} \tag{4.4}$$

Notation remark: The explicit dependence of state and input variables on time is omitted for convenience

Once the variable t_f is added into the 3-DoF vehicle dynamic model, it can be used as an optimization variable in the optimal guidance problem, and therefore its value can be decided by the optimization solver.

This modification causes the mathematical model to become nonlinear.

4.2.2 Optimal Control Problem Formulation

The free-final-time optimal guidance is formulated using the classic optimal control problem scheme in continuous-time as follows:

Cost Function:

$$\min_{T(\cdot), t_f} J = \left(\int_{t_0}^{t_f} \lambda_0 \cdot t_f + \lambda_1 \cdot \|T(t)\| + \lambda_2 \cdot \|\dot{T}(t)\|^2 dt \right)$$

subject to:

Boundary Conditions:

$$\begin{aligned} p(t_0) &= p_0 & \|p(t_f) - p_f\| &\leq p_{tol} \\ v(t_0) &= v_0 & \|v(t_f) - v_f\| &\leq v_{tol} \end{aligned}$$

3-DoF Vehicle Dynamics:

$$\begin{aligned} \dot{p}(t) &= v(t) \cdot t_f \\ \dot{v}(t) &= \left(\frac{1}{m} \cdot T(t) + g \right) \cdot t_f \end{aligned}$$

Ascent/Descent Glideslope Constraints:

$$\begin{aligned} (1, 0, 0) \cdot (p(t) - p(t_0)) &\geq \tan(\gamma_{gs}) \cdot \|(0, 1, 1) \cdot (p(t) - p(t_0))\| \\ (1, 0, 0) \cdot (p(t) - p(t_f)) &\geq \tan(\gamma_{gs}) \cdot \|(0, 1, 1) \cdot (p(t) - p(t_f))\| \end{aligned}$$

State Constraints:

$$\|v(t)\| \leq v_{max}$$

Thrust Constraints:

$$\begin{aligned} T_{min} &\leq \|T(t)\| \leq T_{max} \\ \dot{T}_{min} &\leq \|\dot{T}(t)\| \leq \dot{T}_{max} \\ (1, 0, 0) \cdot T(t) &\geq \cos(\theta_{max}) \cdot \|T(t)\| \end{aligned}$$

Free-final-time Constraints:

$$t_f \geq 0$$

where $p(t) = [p_x(t), p_y(t), p_z(t)]$ is the linear position in the 3D space, $v(t) = [v_x(t), v_y(t), v_z(t)]$ is the linear velocity vector and $T(t) = [T_x(t), T_y(t), T_z(t)]$ represents the thrust vector input.

It is important to remark some key aspects of the previous formulation:

- The optimization problem is meant to be solved through the manipulation of the optimization variables $T(\cdot)$ (thrust vector) and t_f (trajectory total elapsed time).
- The cost function is intended to minimize the total time elapsed by the trajectory through the first term of it. In addition, the second and third terms of the cost function are added to minimize the thrust input effort as well as to penalize large variations of it. The different terms of the cost function can be weighted using the scalars λ_0 , λ_1 and λ_2 , with the objective of prioritizing one of the optimality criterion over the other, as well as for reaching a trade-off between them. Furthermore, due to the fact that the terms of the cost function are expressed in different units, they should be normalized.
- The boundary conditions constraint the problem by imposing an initial condition for the state of the vehicle, as well as a soft final condition defined as a close to the target state neighborhood, whose precision can be increased or decreased by tuning the p_{tol} and v_{tol} parameters.
- The ascent/descent glideslope 4.1 expressions constraint the shape of the trajectory by establishing a relationship between the vertical altitude of the vehicle and its horizontal distance to the target point, which seeks to avoid off-centered trajectories.

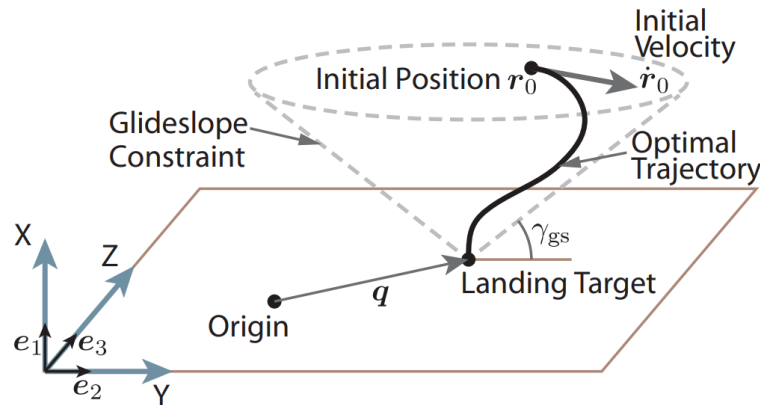


Figure 4.1: Glideslope constraint visualization from [5]

- The state constraint prevents the vehicle from gaining too much linear velocity.
- Thrust constraints have the objective of imposing the input operation range, by defining a lower magnitude bound for the thrust vector T_{min} and an upper magnitude bound for the thrust vector T_{max} . In addition, a maximum vehicle tilt angle (θ_{max}) constraint is imposed

by limiting the inclination of the thrust vector.

- The total trajectory elapsed time t_f is constrained to have a duration greater or equal than zero, in order to avoid mathematically feasible but physically unrealistic solutions to the guidance problem.

The continuous-time guidance formulation used in this master thesis is inspired by state of the art works [1, 13, 16] done in the GNC field, but has two slight modifications: a term accounting for the final trajectory time is explicitly added to the cost function, while removing a term accounting for the slackness of the maximum velocity constraint.

In order to enable software implementations of the guidance problem, the continuous-time formulation is discretized using Euler's method for the sake of simplicity. By doing so, the solution of the discrete free-final-time optimal guidance problem is expected to be a set of N waypoints that represent a discretized trajectory. As a consequence, each of the waypoints will represent a timestamped 3-DoF state of the vehicle (position and velocity) and thrust vector (T_x, T_y, T_z) that will be used as a setpoint for lower tracking control layers.

To do so, the time-scaled interval $[0,1]$ is discretized in $N = 100$ samples, producing a fixed sampling time of $T_{fixed} = \frac{1}{N}$ seconds. Once the guidance problem is solved and the value of the trajectory elapsed time t_f is found, the trajectory is time-scaled back to the interval $[0, t_f]$ where the time step is scaled to $T_s = t_f \cdot T_{fixed}$.

Finally, the initial continuous-time guidance expressions are discretized, resulting in the following formulation:

Cost Function:

$$\min_{x(\cdot), T(\cdot), t_f} J = \left(\sum_{k=0}^N \lambda_0 \cdot t_f + \lambda_1 \cdot \|T(k)\| + \lambda_2 \cdot \left\| \frac{T(k+1) - T(k)}{T_{sfixed}} \right\|^2 \right)$$

subject to:

Boundary Conditions:

$$\begin{aligned} p(0) &= p_0 & \|p(N+1) - p_f\| &\leq p_{tol} \\ v(0) &= v_0 & \|v(N+1) - v_f\| &\leq v_{tol} \end{aligned}$$

3-DoF Vehicle Dynamics:

$$\begin{aligned} p(k+1) &= p(k) + v(k) \cdot t_f \cdot T_{sfixed} \\ v(k+1) &= v(k) + \left(\frac{1}{m} \cdot T(k) + g \right) \cdot t_f \cdot T_{sfixed} \end{aligned}$$

Ascent/Descent Glideslope Constraints:

$$\begin{aligned} (1, 0, 0) \cdot (p(k) - p(t_0)) &\geq \tan(\gamma_{gs}) \cdot \|(0, 1, 1) \cdot (p(k) - p(t_0))\| \\ (1, 0, 0) \cdot (p(k) - p(N+1)) &\geq \tan(\gamma_{gs}) \cdot \|(0, 1, 1) \cdot (p(k) - p(N+1))\| \end{aligned}$$

State Constraints:

$$\|v(k)\| \leq v_{max}$$

Thrust Constraints:

$$\begin{aligned} T_{min} &\leq \|T(k)\| \leq T_{max} \\ \dot{T}_{min} &\leq \left\| \frac{T(k+1) - T(k)}{T_{sfixed}} \right\| \leq \dot{T}_{max} \\ (1, 0, 0) \cdot T(k) &\geq \cos(\theta_{max}) \cdot \|T(k)\| \end{aligned}$$

Free Final Time Constraints:

$$t_f \geq 0$$



where $p(k) = [p_x(k), p_y(k), p_z(k)]$ is the linear position in the 3D space, $v(k) = [v_x(k), v_y(k), v_z(k)]$ is the linear velocity vector and $T(k) = [T_x(k), T_y(k), T_z(k)]$ represents the thrust vector input.

The guidance problem is solved once for each vehicle's launch/landing maneuver, and its solution consists of an optimal trajectory formed by N discrete waypoints, as well as an optimal time step T_s that is directly derived from the total flight time t_f .

4.3 Nonlinear Model Predictive Position Control

Once the guidance subsystem of the rocket has been designed, a control method is required in order to provide the rocket with the actual capability to track a given trajectory. As a consequence, the main objective of the control subsystem is to, given a optimal guidance trajectory, generate a sequence of thrust vector control inputs that manipulate the behavior of the vehicle in such a way that it follows the provided path.

In contrast to the guidance subsystem, the control subsystem is intended to use the 6-DoF vehicle's model to achieve a more accurate and closer-to-reality response, since this model is derived to represent the dynamics of the rocket in a more complete manner, by not only considering its translational motion, but also modelling its rotational motion. From an intuitive point of view, the 3-DoF model is sufficiently precise for performing planning tasks, but when dealing with the rocket actual motion control, the rotational motion of the vehicle cannot be neglected and the use of the 6-DoF model is required in order to implement a realistic controller.

As mentioned in the previous subsection, the output of the guidance subsystem is an optimal trajectory expressed as a sequence of N timestamped waypoints of the form:

$$r(k) = \{ (p(0), v(0), t = 0 \cdot T_s), (p(1), v(1), t = 1 \cdot T_s), \dots, (p(N), v(N), t = N \cdot T_s) \}, \quad (4.7)$$

where each of the N trajectory waypoints indicates a reference value for the linear position and linear velocity for the control subsystem to achieve at a certain discrete time instant.

Due to the fact that the 6-DoF vehicle's model consider more dynamics and state variables than the 3-DoF vehicle's model, the trajectory waypoints partially describe the desired state to achieve by the control subsystem, only fixing the linear position and velocity, but leaving free the angular velocity and quaternion attitude of the vehicle. As no reference is provided for these additional state variables, their evolution along the trajectory is not desired to evolve in a predefined manner, but this does not mean that its dynamics are not subject to any additional required constraints, as the ones formulated in (4.9).

Given the complexity of the 6-DoF nonlinear state space rocket's model, the necessity of both

imposing several constraints in the control process and implementing a multivariable control algorithm, a nonlinear model-based predictive control strategy has been selected for implementing the control subsystem. This decision is also reinforced by other related state of the art works done in the area of rockets GNC and represent a perfect opportunity for leveraging work done in previous research projects as [1, 23].

Similarly to conventional model predictive control (MPC) strategies, nonlinear model predictive control (NMPC) [24] is used to calculate adequate control actions along a finite-time prediction horizon by using nonlinear constrained optimization and nonlinear model-based predictions. MPC and NMPC formulations are usually rather similar to the one used for nonlinear optimal control problems. While this latter type of problems are only solved once, and provide a sequence of control actions to be applied in an open-loop manner (N)MPC problems are solved in a repeated manner, whereby only the first control action from each of the calculated sequences is applied. Then, its effect on the system is measured, and a new solution for the optimal control problem is computed by updating the prediction horizon and initial conditions, endowing the process with closed-loop decision making capabilities.

A relevant disadvantage to highlight in relation to NMPC is that, while the linear MPC formulation can be formulated as a convex optimization problem where stability and convergence conditions are well defined, the nonlinear formulation of the same problem may produce non-convex optimization problems in which stability and convergence conditions are not clear anymore, and therefore is still an open problem in the state of art [25].

In general terms, model based predictive control strategies require two type of models to work as expected:

- A prediction model, used to compute the approximated evolution of the system along the prediction horizon with the objective of calculating the optimal control actions to take, based of the effect of those in the dynamics.
- A simulation model, used to compute the actual effect of the optimal control actions on the system. This model is represented by the real-life plant when the control strategy is implemented in an experimental setup, but is usually substituted by an augmented model which contemplates additional unmodeled dynamics, disturbances or noise affecting the prediction model when the control strategy is implemented in a simulation setup.

Due to the fact that the activities of this thesis are only carried in a simulation setup, the two previously mentioned models are described hereafter.

4.3.1 6-DoF Control-Oriented Dynamic Model

The control-oriented model is intended to work as a prediction model for the NMPC, by providing it with the capability of computing the approximated evolution of the vehicle's dynamics

along the prediction horizon.

The control-oriented model used to fulfill this requirement is the previously described 6-DoF rocket dynamic model formulated in Section 3.4.4.

4.3.2 6-DoF Simulation-Oriented Dynamic Model

The simulation-oriented model is intended to work as a closer-to-reality vehicle's representation that may contain additional characteristics that are not modeled in the control oriented model.

Due to the fact that all the relevant dynamics of the vehicle are already considered within the control-oriented model, an augmented uncertainty model is proposed in order to simulate the possible linear velocity and angular velocity IMU state measuring error, which is a practical case that is often found in experimental setups.

In order to do so, the linear velocity and angular velocity states from 3.11 are altered by an additive Gaussian noise ν as follows:

$$\nu = N(\mu, \Sigma) = N(0, 0.1) \quad (4.8)$$

4.3.3 Final Nonlinear Model Predictive Position Control Problem Formulation

The nonlinear model predictive position control problem is mathematically formulated as follows, using the classical optimal control problem scheme:

Cost Function:

$$\min_{T(\cdot)} J = \left(\int_{t_0}^{H_p} (x_{p,v}(t) - r(t)) \cdot Q \cdot (r(t) - x_{p,v}(t))^T + T(t) \cdot R \cdot T(t)^T + \omega(t) \cdot S \cdot \omega(t)^T dt \right)$$

subject to:

Boundary Conditions:

$$\begin{aligned} p(t_0) &= p_0 \\ v(t_0) &= v_0 \\ q(t_0) &= q_0 & \|q(t_f) - q_f\| &\leq q_{tol} \\ \omega(t_0) &= \omega_0 & \|\omega(t_f) - \omega_f\| &\leq \omega_{tol} \end{aligned}$$

Trajectory Tracking Constraints:

$$\|x_{p,v}(t) - r(t)\| \leq \text{waypoint}_{tol}$$

6-DoF Vehicle Dynamics:

$$\begin{aligned} \dot{p}(t) &= v(t) \\ \dot{v}(t) &= R(q) \cdot \frac{T(t)}{m} + g \\ \dot{q}(t) &= \frac{1}{2} \cdot \Omega(\omega) \cdot q(t) \\ \dot{\omega}(t) &= \frac{1}{J} \cdot (T(t) \times r_{CoM} - \omega(t) \times (J \cdot \omega(t))) \end{aligned}$$

State Constraints:

$$\|q(t)\| = 1$$

Thrust Constraints:

$$\begin{aligned} T_{min} &\leq \|T(t)\| \leq T_{max} \\ -T_x(t) \cdot \tan(\delta_{max}) &\leq T_y(t) \leq T_x(t) \cdot \tan(\delta_{max}) \\ -T_x(t) \cdot \tan(\delta_{max}) &\leq T_z(t) \leq T_x(t) \cdot \tan(\delta_{max}) \end{aligned}$$



where $p(t) = [p_x(t), p_y(t), p_z(t)]$ is the linear position in the 3D space, $v(t) = [v_x(t), v_y(t), v_z(t)]$ is the linear velocity vector, $x_{p,v}(t) = [p(t), v(t)]$ is the partial state formed by linear position and velocity, and $T(t) = [T_x(t), T_y(t), T_z(t)]$ represents the thrust vector input.

The key aspects of the previous formulation are the following:

- The optimization problem is meant to be solved in the range $t \in [t_0, t_0 + Hp]$ through the manipulation of the thrust vector $T(\cdot)$ optimization variable. Differently from the guidance formulation, the problem is solved for a certain time range within the full trajectory, defined by the prediction horizon Hp . This value, which should be fine-tuned to enhance the predictive capabilities of the controller, is generally desired to be smaller than the full trajectory duration t_f .
- The first term of cost function is intended to minimize the tracking error between the linear position and velocity described by the reference trajectory and the current state of the vehicle. Additionally, the second and third terms are intended to minimize the thrust control effort and the angular velocity vector of the rocket, respectively. These terms are weighted by the matrices Q , R and S , respectively.
- The boundary conditions constrain the problem by imposing an initial condition for the vehicle state, as well as a soft final condition for the free state variables that are not meant to follow the reference trajectory. The final condition for these states is such that the rocket is not rotating around any of its axes and is pointing up straight.
- The trajectory constraint aims to keep the distance between the planned and executed trajectory within a tolerance range that can be tuned.
- The state constraint for the attitude quaternion vector establishes that its norm should be unitary, in order to guarantee its consistency when representing the 3D orientation of the vehicle.
- Thrust constraints have the objective of restricting the input operation range, by defining lower and upper magnitude bounds for the thrust vector, given by T_{min} and T_{max} , respectively. In addition, a maximum thruster gimbal tilt angle (δ_{max}) constraint is imposed in order to represent the physical limits of the actuator in the YZ plane.

As it was done for the guidance subsystem, and due to the fact that the reference trajectory is expressed in discrete time, the nonlinear model predictive position controller formulation is discretized using Euler's method in order to allow its software implementation:

Cost Function:

$$\min_{x(\cdot), T(\cdot)} J = \left(\sum_{k=0}^{Hp} (x_{p,v}(k) - r(k)) \cdot Q \cdot (r(k) - x_{p,v}(k))^T + T(k) \cdot R \cdot T(k)^T + \omega(k) \cdot S \cdot \omega(k)^T \right)$$

subject to:

Boundary Conditions:

$$p(0) = p_0$$

$$v(0) = v_0$$

$$q(0) = q_0 \quad \|q(Hp + 1) - q_f\| \leq q_{tol}$$

$$\omega(0) = q_0 \quad \|\omega(Hp + 1) - \omega_f\| \leq \omega_{tol}$$

Trajectory Tracking Constraints:

$$\|x_{p,v}(k) - r(k)\| \leq waypoint_{tol}$$

6-DoF Vehicle Dynamics:

$$p(k + 1) = p(k) + v(k) \cdot T_s$$

$$v(k + 1) = v(k) + \left(R(q) \cdot \frac{T(k)}{m} + g \right) \cdot T_s$$

$$q(k + 1) = q(k) + \left(\frac{1}{2} \cdot \Omega(\omega) \cdot q(k) \right) \cdot T_s$$

$$\omega(k + 1) = \omega(k) + \left(\frac{1}{J} \cdot (T(k) \times r_{CoM} - \omega(k) \times (J \cdot \omega(k))) \right) \cdot T_s$$

State Constraints:

$$\|q(k)\| = 1$$

Thrust Constraints:

$$T_{min} \leq \|T(k)\| \leq T_{max}$$

$$-T_x(k) \cdot \tan(\delta_{max}) \leq T_y(k) \leq T_x(k) \cdot \tan(\delta_{max})$$

$$-T_x(k) \cdot \tan(\delta_{max}) \leq T_z(k) \leq T_x(k) \cdot \tan(\delta_{max})$$



where $p(k) = [p_x(k), p_y(k), p_z(k)]$ is the linear position in the 3D space, $v(k) = [v_x(k), v_y(k), v_z(k)]$ is the linear velocity vector, $x_{p,v}(k) = [p(k), v(k)]$ is the partial state formed by linear position and velocity, and $T(k) = [T_x(k), T_y(k), T_z(k)]$ represents the thrust vector input.

The position control problem is solved as many times as the number of waypoints present in the reference guidance trajectory, and its solution consists in a sequence of thrust vector commands for the vehicle's main engine to execute.

4.4 Implementation using Matlab & CasADi

The previously described theoretical concepts have been implemented in a simulation setup using Matlab and CasADi software tools, with the objective of creating several simulation cases in which the behavior of the vehicle's models and control strategies can be put into practice.

MATLAB [26] (short for MATrix LABoratory) is a high-level programming language and computing environment developed by MathWorks. It is widely used for numerical computation, data analysis, algorithm development, visualization, and simulation. The use case subject of this work is perfectly matched by this tool.

Furthermore, CasADi is an open-source software tool for symbolic computation, automatic differentiation (AD), and numerical optimization, particularly intended for solving optimal control and nonlinear programming problems. In addition, CasADi includes a very interesting toolbox that is highly useful for this work, called Opti Stack, which is a collection of tools built on top of CasADi, and which are designed to simplify the formulation and solution of optimization problems in MATLAB. It provides a unified interface for various solvers and libraries, allowing users to define and solve problems formulated using linear programming (LP), nonlinear programming (NLP), quadratic programming (QP), mixed-integer programming (MIP), etc.



(a) Matlab Logo [26]



(b) CasADi Logo [27]

Due to the fact that both optimal guidance and nonlinear model predictive position control represent nonlinear and nonconvex problems, the Interior Point OPTimizer (IPOPT) nonlinear programming optimization solver has been found to be the best open source option for this work. IPOPT is an open-source software package for solving large-scale nonlinear optimization problems, and constitutes a popular solver used for problems involving continuous variables with nonlinear constraints and objective functions. IPOPT is based on the interior-point method, a class of algorithms that are particularly effective for optimization problems that involve non-

linearities in both the objective function and the constraints.

However, it is important to remark that other commercial optimization solver alternatives exist within the state of the art of the aerospace GNC field, as for example, ForcesPro by Embotech, which has been proved to offer nice results and performance as in previous works [1, 13, 28].

The specific details involving the implementation of both the optimal guidance and position control subsystems are described hereafter.

4.4.1 3-DoF Free-Final-Time Optimal Guidance

The free-final-time optimal guidance has been implemented in a Matlab script that makes use of the CasADi Opti Stack toolbox for solving the optimal control problem through IPOPT solver. The following algorithmic pseudocode summarizes its actions:

Algorithm 1 Free-Final-Time Optimal Guidance - Open Loop

Input: Initial Vehicle 3-DoF State $[p_0, v_0]$, Target Vehicle 3-Dof State $[p_f, v_f]$

Initialize model parameters
 Initialize boundary conditions
 Initialize optimization problem parameters
 Initialize CasADi Opti Stack variables
 Initialize symbolic Vehicle Dynamics

for $k = 1 : N$ **do**

- Symbolically Compute next state imposing the discrete vehicle dynamics in Section 3.3
- Symbolically Compute cost function defined in Section 4.2.2 for current iteration k
- Symbolically Include constraints defined in Section 4.2.2 for current iteration k

end for

Solve the optimization problem by minimizing the cost function using IPOPT
 Record the obtained solution

Output: Trajectory $r(k) = \{waypoint_0, \dots, waypoint_N\}$, Trajectory Time t_f

4.4.2 6-DoF nonlinear Model Predictive Position Control

In a similar manner as in the previous subsection, the nonlinear model predictive position control has been implemented in a Matlab script that makes use of the CasADi Opti Stack toolbox for solving the optimal control problem through IPOPT solver.

This scripts simulates the whole NMPC control loop by using two nested loops: the outer loop iterates along waypoints of the reference trajectory, and for each of them, the inner loop solves the optimal control problem for the next H_p steps, provides a solution, applies it and moves one step forward in the simulation by updating the initial conditions of the next problem iter-

ation. This nested loop configuration is represented by the existence of two time horizons: the trajectory time horizon, which is defined from the initial point of the planned trajectory to its final point; and the moving NMPC prediction horizon, that, as the simulation evolves, moves along all the elements of the global time horizon. In order to trace both loops, two independent iteration variables are defined, and additionally, a third iteration variable is introduced in order to have a global reference of the trajectory along the prediction horizon.

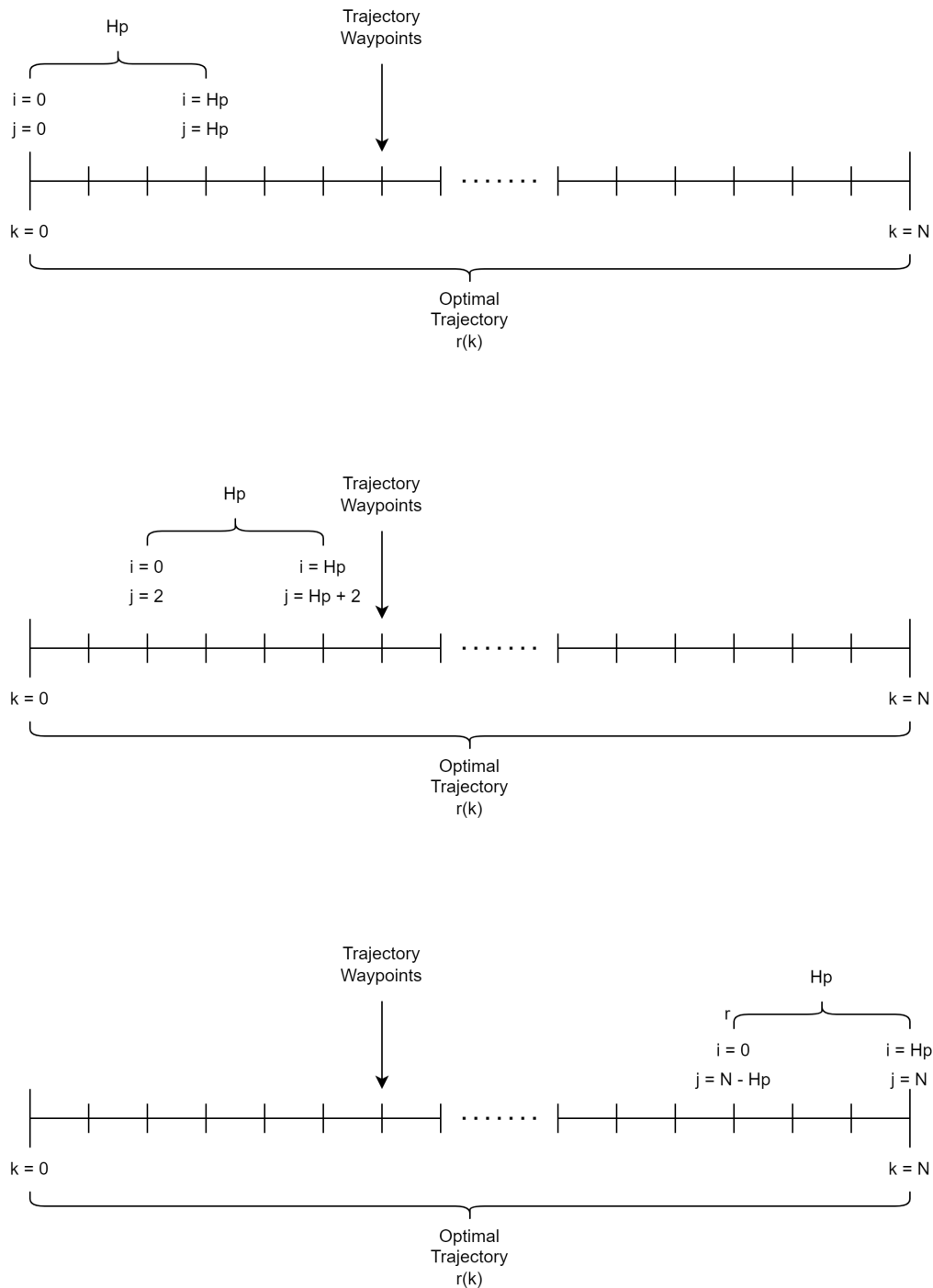


Figure 4.3: Closed Loop Simulation Horizon Structure

An important issue that should be remarked emerges when, given the general case where the trajectory is formed by N points and the NMPC prediction horizon is H_p , at the $N - H_p$ iteration, the last waypoint of the trajectory is reached by the inner loop. From that moment on, for example at the $N - H_p + 1$ iteration, the reference trajectory is extended by repeating its very last element, which represents the target final position, in order to fulfil the prediction horizon of the NMPC, which at this point of the simulation exceeds the length of the trajectory.

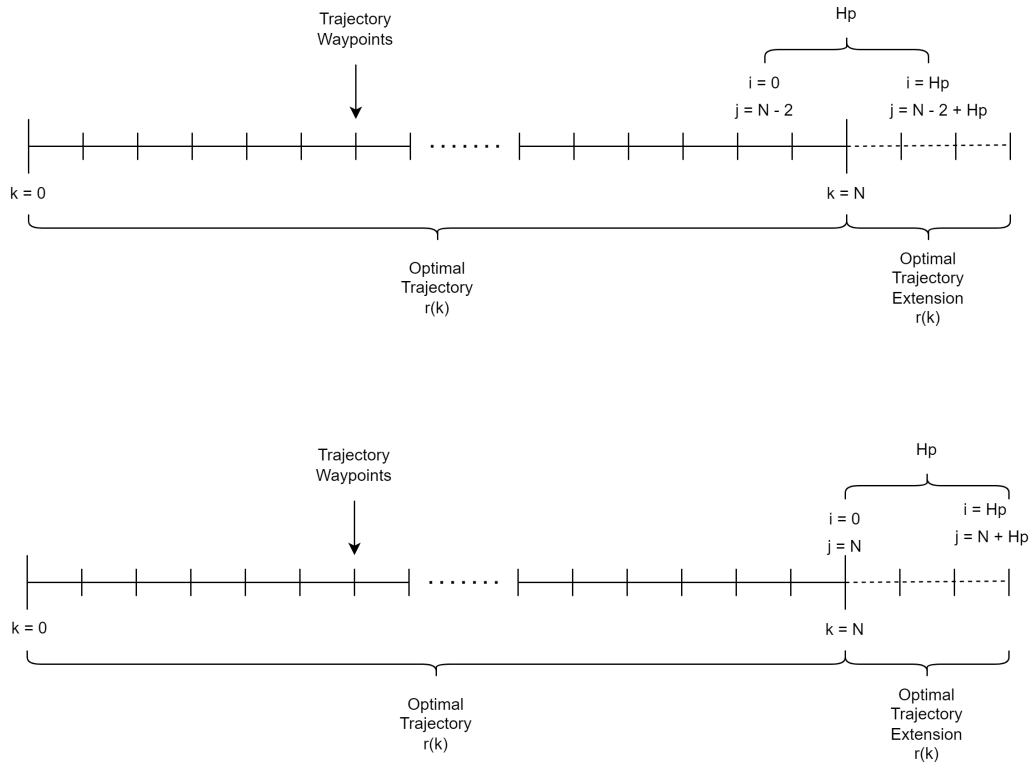


Figure 4.4: Closed Loop Simulation Extended Horizon Structure

The integration between the free final guidance algorithm and the nonlinear model predictive position controller is done as follows:

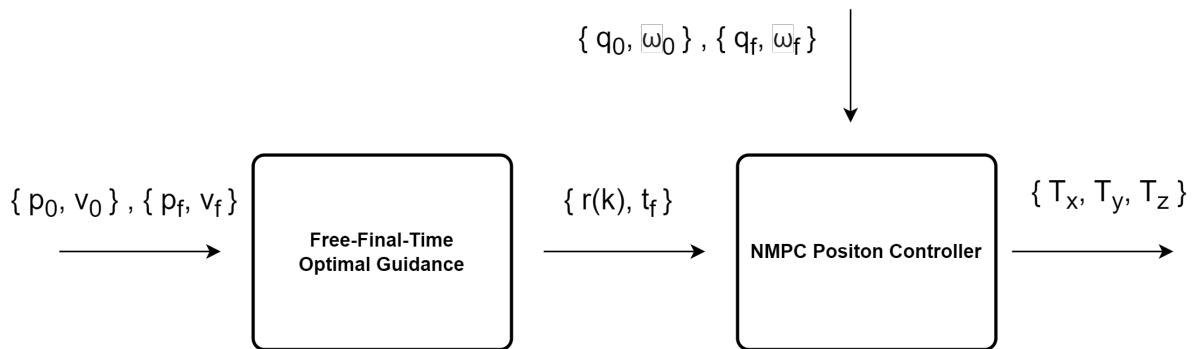


Figure 4.5: Integration diagram between guidance and control subsystems

First of all, the guidance subsystem receives the initial and target conditions, calculates the reference trajectory and its time duration, which is the input for the NMPC controller that actually performs the reference tracking by executing a sequence of adequately calculated thrust vector commands.

The following algorithmic pseudocode summarizes the previously described implementation:

Algorithm 2 Nonlinear Model Predictive Position Control - Closed Loop

Input: Initial Vehicle 6-DoF State $[r_p(0), r_v(0), q_0, \omega_0]$, Target Vehicle 6-DoF State $[r_p(N), r_v(N), q_f, \omega_f]$, Reference Trajectory $r(k)$

Initialize model parameters

Initialize boundary conditions

Initialize optimization problem parameters

Initialize symbolic Vehicle Dynamics

for $k = 1 : N$ **do**

- Initialize CasADi Opti Stack variables

for $i = 1 : H_p$ **do**

- Compute the value of the global trajectory pointing index as $j = k + i - 1$

- Symbolically Compute next state imposing the discrete vehicle dynamics in Section

4.3.1

- Symbolically Compute cost function defined in Section 4.3.3 for current iteration k

- Symbolically Include constraints defined in Section 4.3.3 for current iteration k

end for

- Solve the optimization problem by minimizing the cost function using IPOPT

- Record the solution, keep the first control action of the sequence

- Simulate the next state of the system with the model in Section 4.3.2 using the current state and the solution

- Update the current state of the system

end for

Output: Sequence of N thrust vector commands applied to the vehicle, Simulated 6-DoF state evolution

Chapter 5

Simulation Results

With the objective of testing the performance of the mathematical models and control strategies described along this thesis, a series of simulations have been designed and executed using the implementation setup detailed in Section [4.4](#).

The different simulation cases are aligned with the main objectives of this work and mainly involve: a initial testing of the mathematical models of the rocket through open-loop maneuvers, followed by the calculation of reference trajectories via the free-final-time optimal guidance algorithm, and finally, the execution of those planned trajectories on closed-loop simulations by using the nonlinear model predictive position controller. These tests specially aim at demonstrating that the rocket is able to achieve actively controlled vertical landing maneuvers using the methods considered in this work.

5.1 Open-Loop Vehicle Dynamics Simulations

Given the fact that landing or divert maneuvers cannot be achieved without actively controlling the motion of the rocket in a closed loop, the most simple test to do at first is an open-loop launch simulation in which the vehicle does not have a reference trajectory to follow.

This test consists in simulating the rocket motion evolution using its 3-DoF model, given an initial condition for its position, velocity and thrust vector. Additionally, when a certain point in time is reached, the thrust vector command is updated. Due to this fact, the rocket is expected to suffer a change in its flight direction:

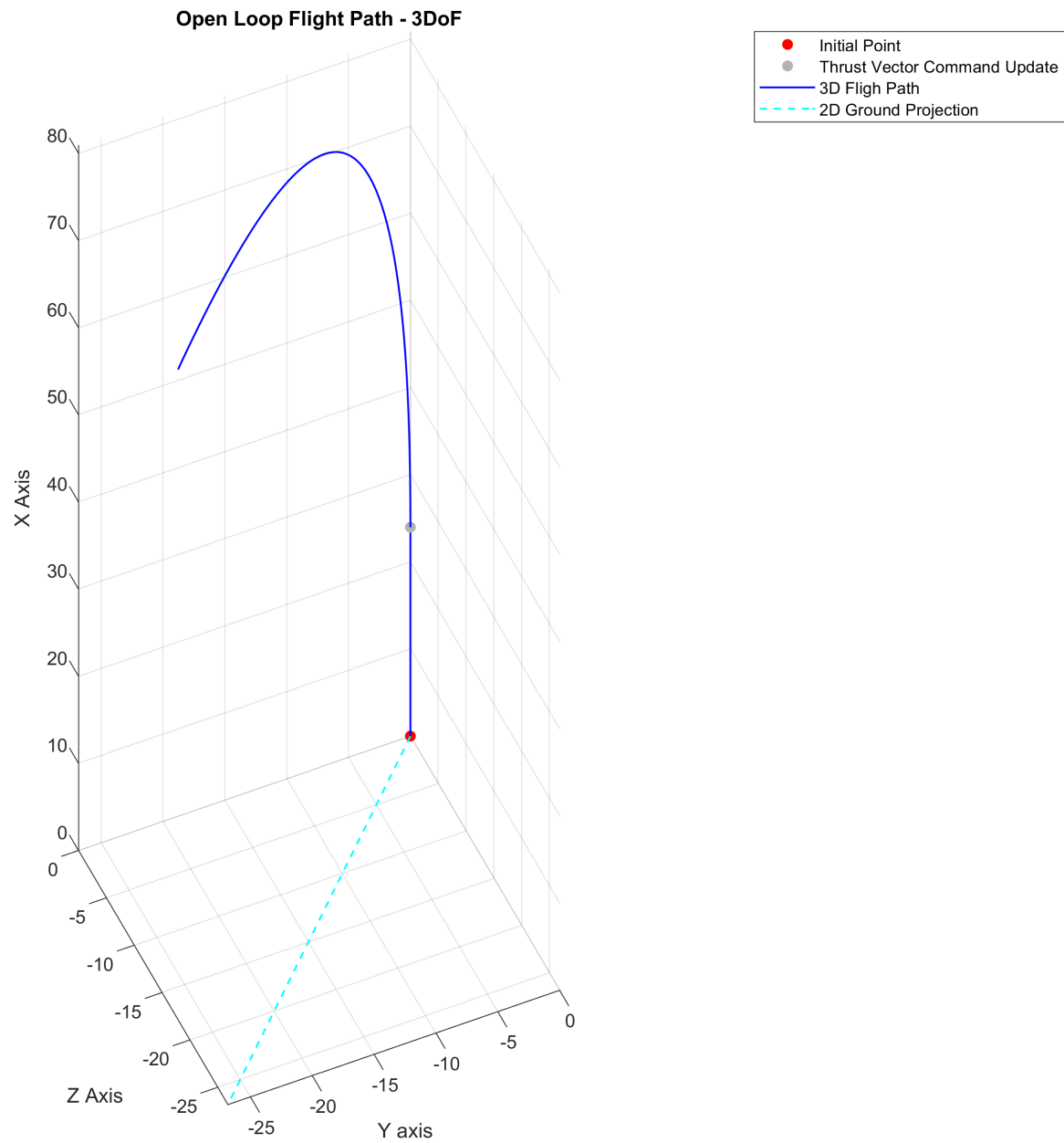
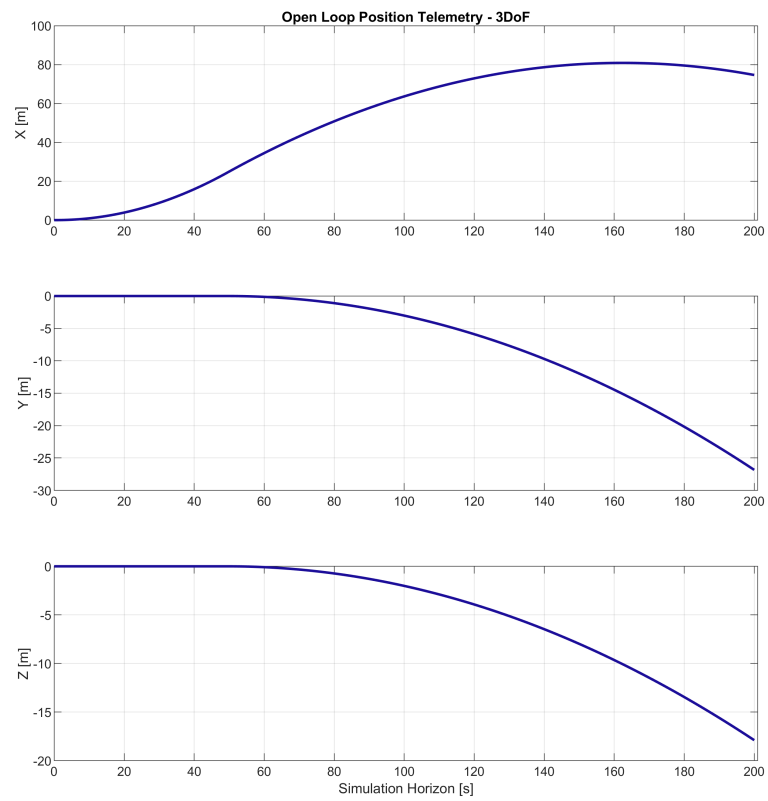
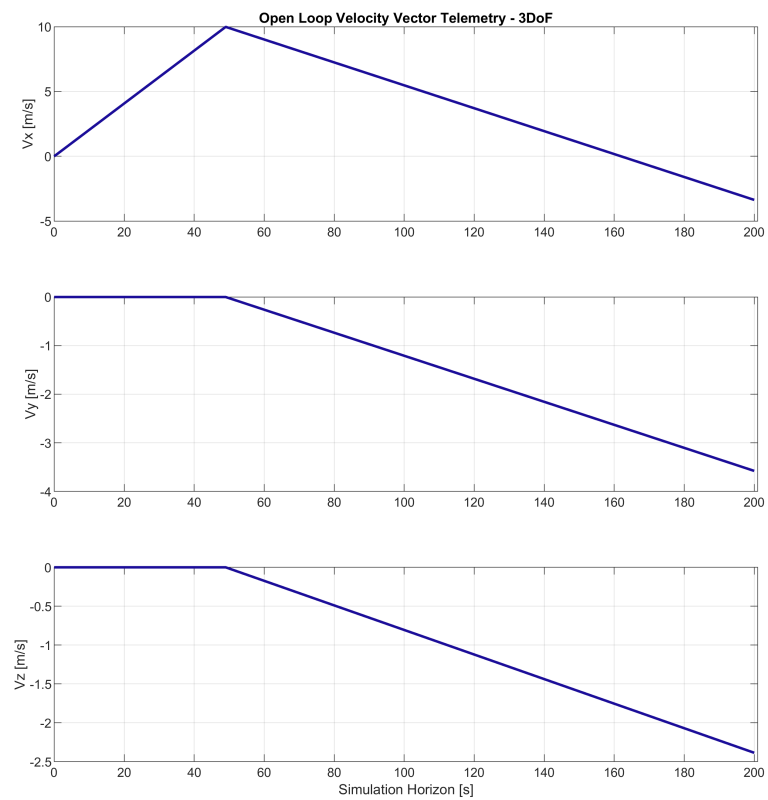


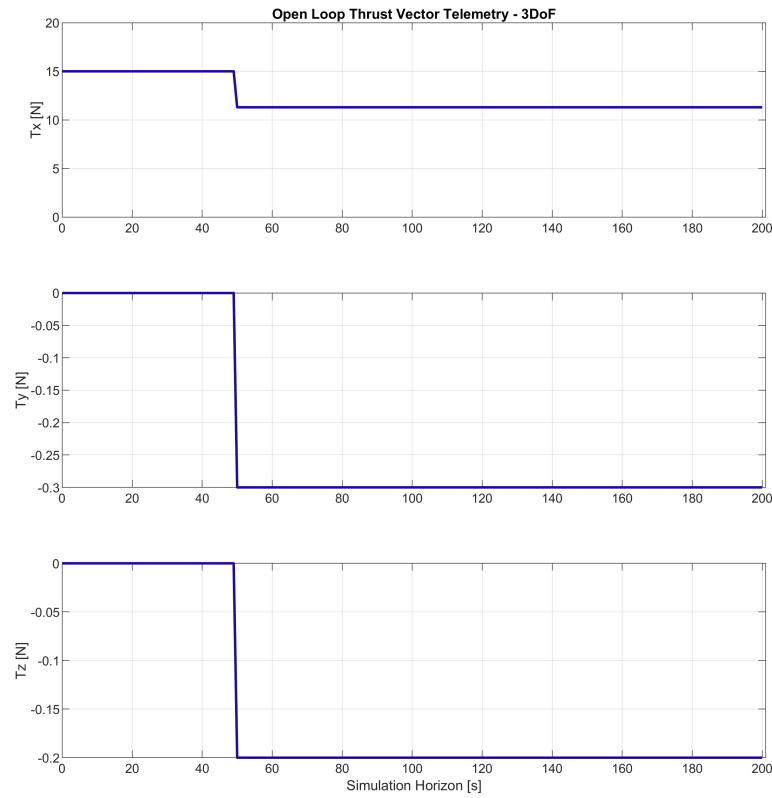
Figure 5.1: 3-DoF Rocket Model Open Loop Flight path for $p_0 = [0, 0, 0]$, $v_0 = [0, 0, 0]$, $T_0 = [15, 0, 0]$, $T_{updated} = [11.3, -0.3, -0.2]$



(a) Linear Position Telemetry



(b) Linear Velocity Vector Telemetry



(c) Thrust Vector Telemetry

Figure 5.2: 3-DoF Rocket Model Open Loop Flight Telemetry

In a similar manner, an open-loop launch simulation case is performed for the 6-DoF vehicle in order to compare the behavior of both models:

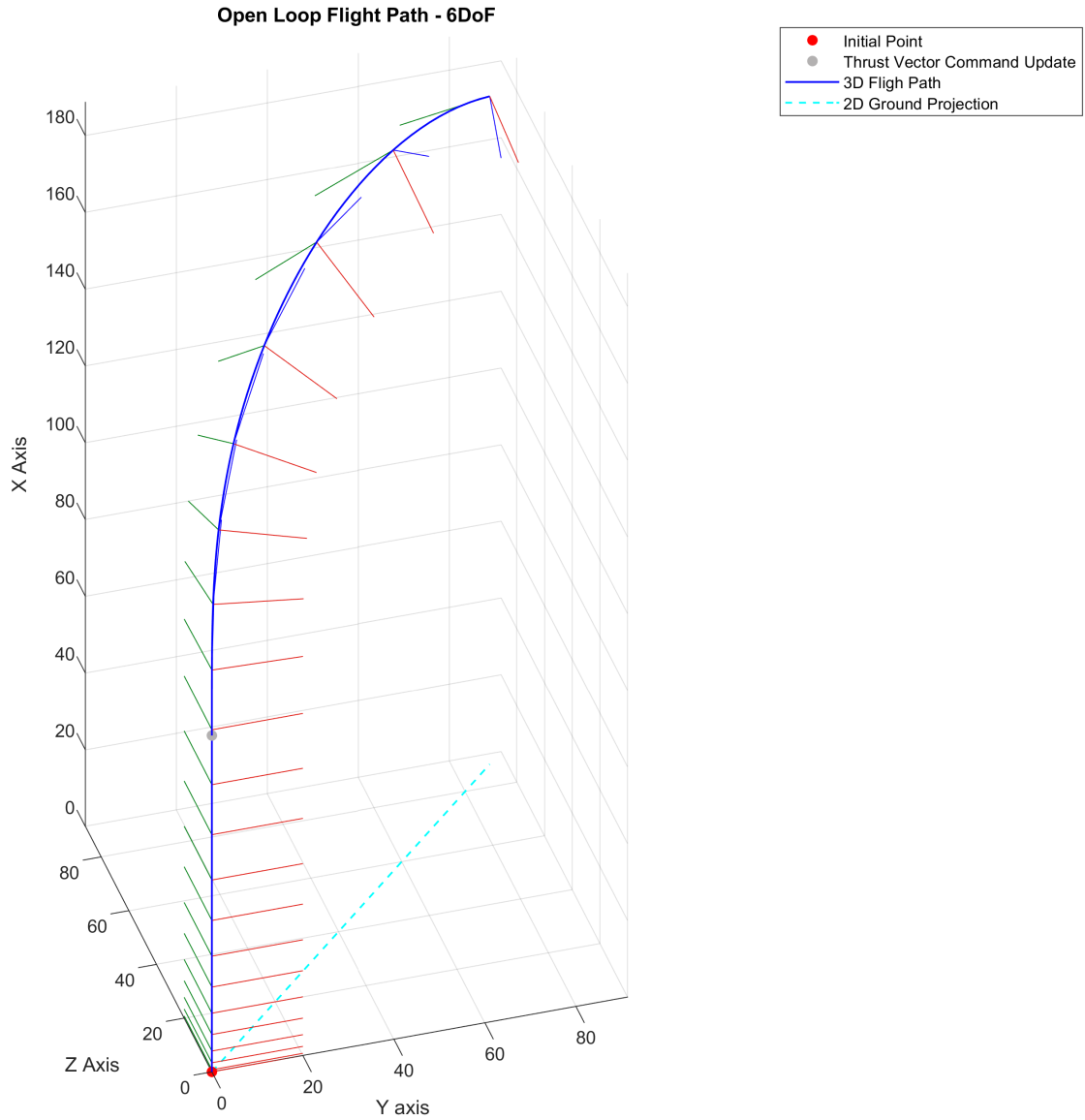
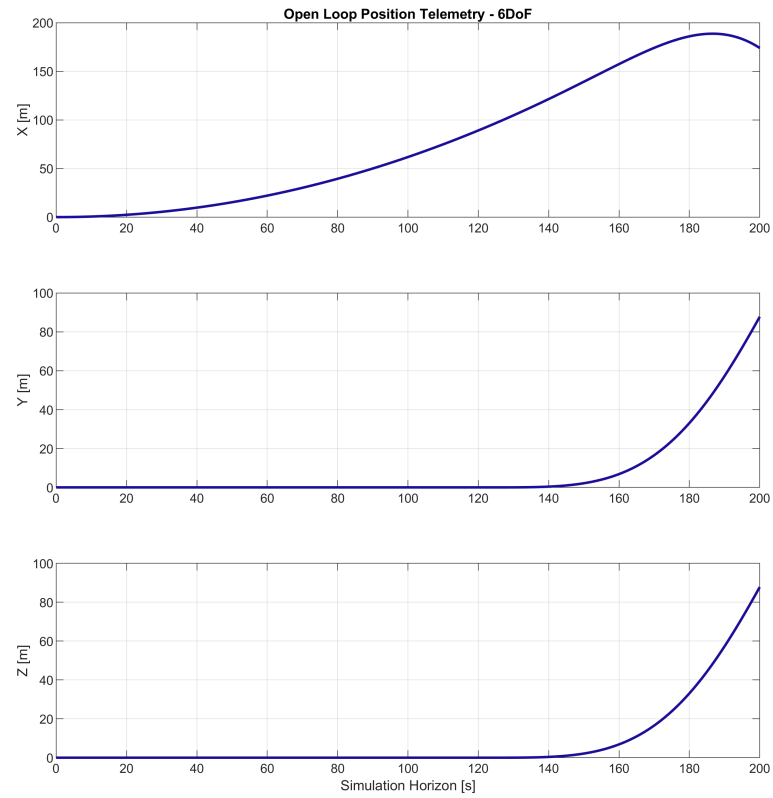
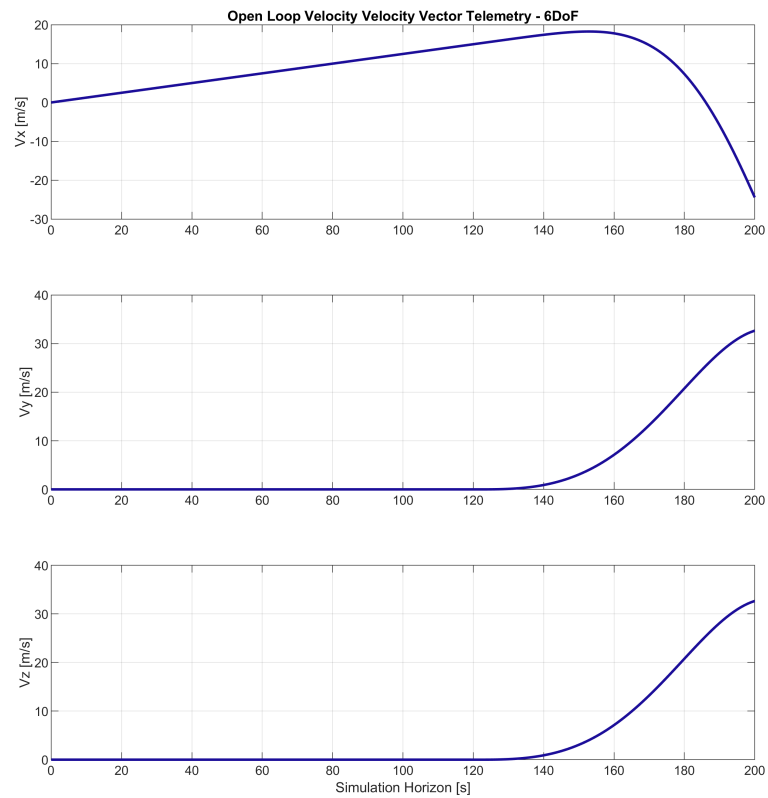


Figure 5.3: 6-DoF Rocket Model Open Loop Flight path for $p_0 = [0, 0, 0]$, $v_0 = [0, 0, 0]$, $q_0 = [1, 0, 0, 0]$, $\omega_0 = [0, 0, 0]$, $T_0 = [14, 0, 0]$, $T_{updated} = [14, -0.015, -0.015]$

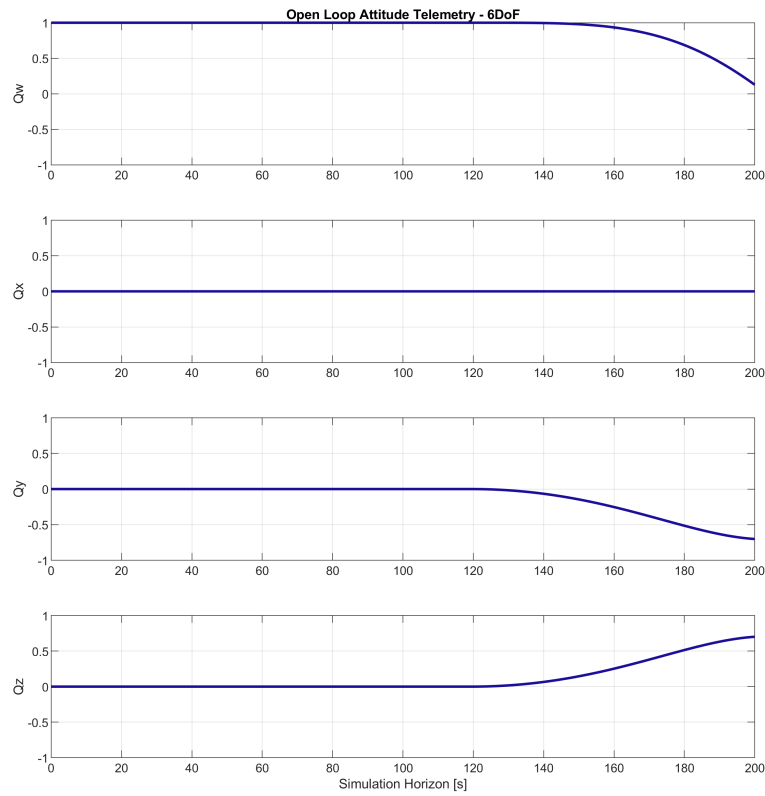
Note: The evolution of the rocket's attitude is represented by the different vehicle frames that appear along the flight path. Initially, the rocket is vertically standing and pointing up, but rotates along the flight path as a result of the TVC actuator.



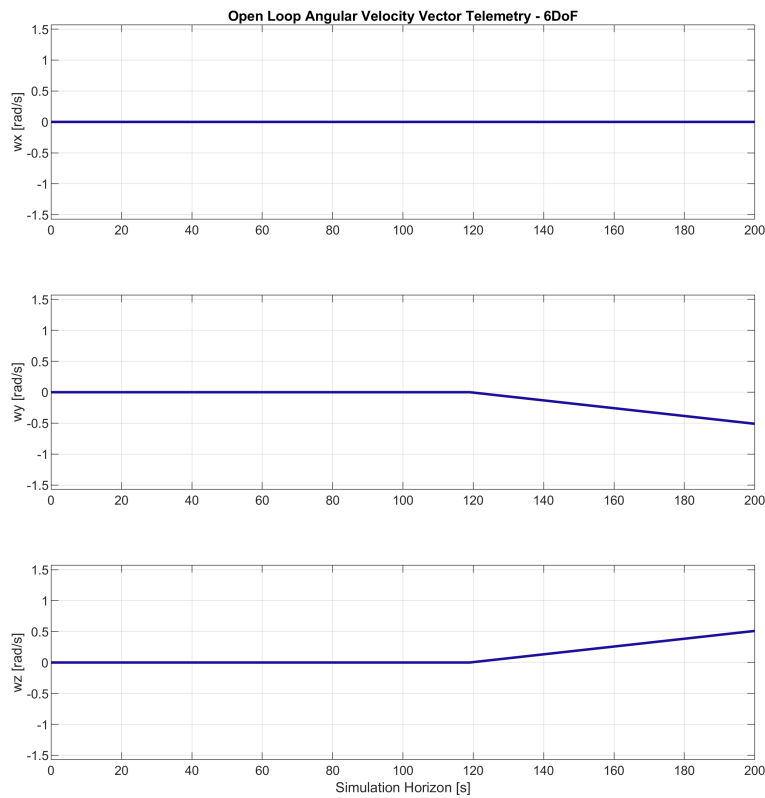
(a) Linear Position Telemetry



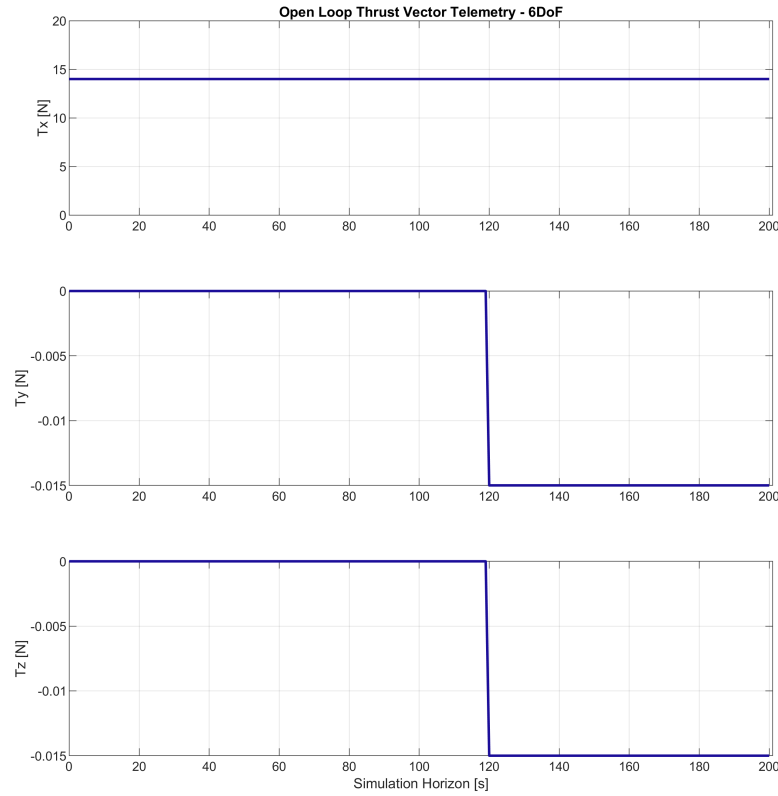
(b) Linear Velocity Vector Telemetry



(c) Attitude Quaternion Telemetry



(d) Angular Velocity Vector Telemetry



(e) Thrust Vector Telemetry

Figure 5.4: 6-DoF Rocket Model Open Loop Flight Telemetry

As it can be seen, when using identical initial conditions and similar thrust vector commands, the dynamic behavior of both models is completely different, being the 6-DoF description much more closer to the real physics of a thrust vector controlled rocket. This is a direct consequence of the complexity difference between both models, as the first one lacks the modelling of the rocket's rotational motion, which dramatically affects its flight direction.

5.2 Optimal Guidance Trajectory Planning Simulations

The free-final-time optimal guidance algorithm is tested by generating a descent soft landing trajectory between an initial point, located mid air, and a target point located in the ground. The initial conditions for the rocket linear velocity are also specified, as well as the target velocity conditions, which are assumed to be such that the rocket is completely at rest. Additionally, the following problem parameters are used along all the tests in this subsection:

Parameter	Value
N	100
p_{tol}	1 m
v_{tol}	3.5 m/s
T_{min}	3 N
T_{max}	20 N
$\dot{T}_{min}$	0 N
$\dot{T}_{max}$	10 N
γ_{gs}	70 °
θ_{max}	5 °
v_{max}	100 m/s

After each of the tests described hereafter, a summary table gathers relevant metrics measuring the performance of the guidance algorithm.

The first test of an optimal guidance for a landing maneuver requires the guidance algorithm to generate a 2D trajectory that connects the initial point $[p_x, p_y, p_z] = [30, 0, 0]$ with the target landing point $[p_x, p_y, p_z] = [0, 8, 0]$. Additionally, the vehicle is assumed to have a initial negative vertical velocity of $[v_x, v_y, v_z] = [-5, 0, 0]$ while its landing velocity should be $[v_x, v_y, v_z] = [0, 0, 0]$:

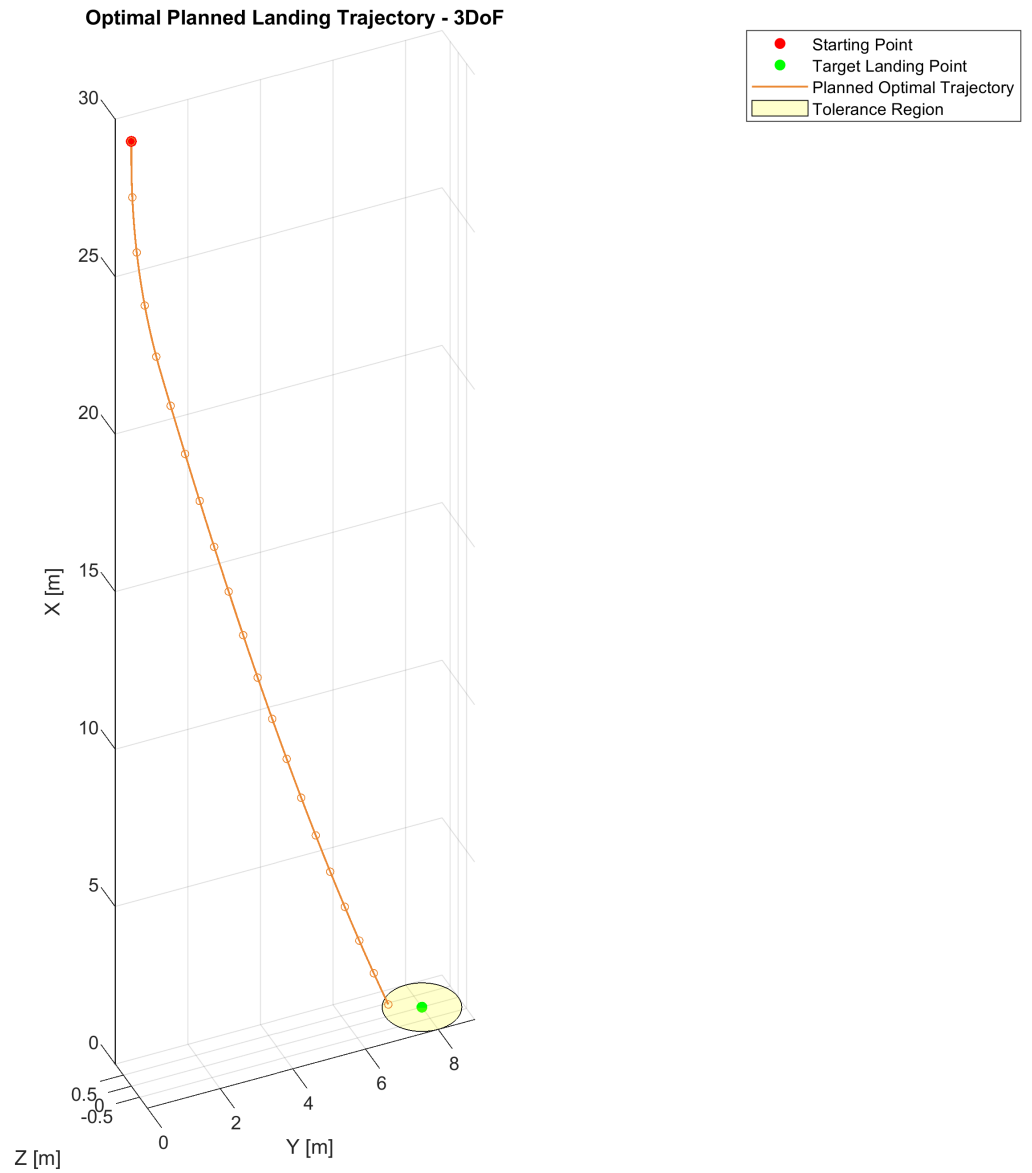
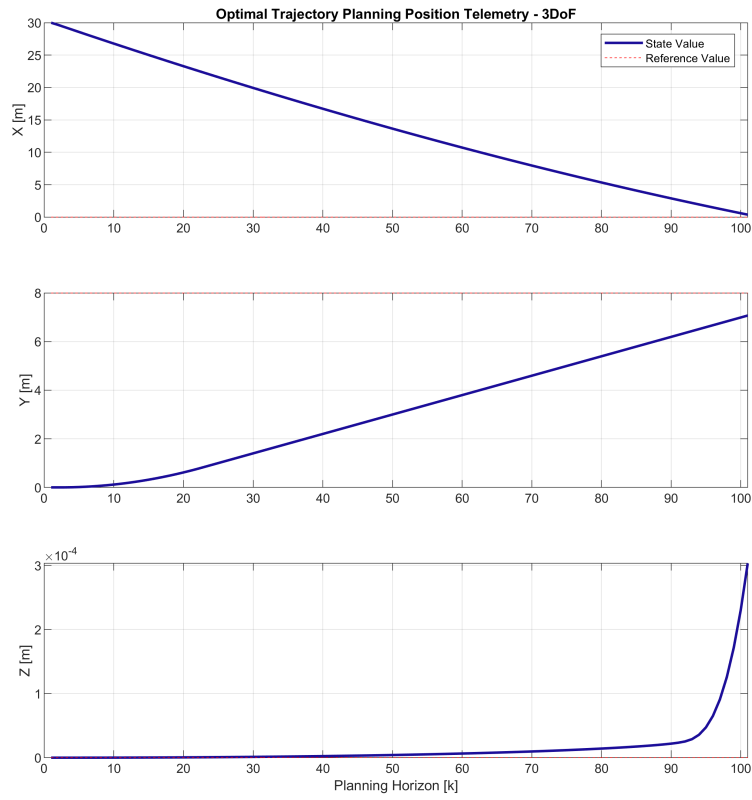
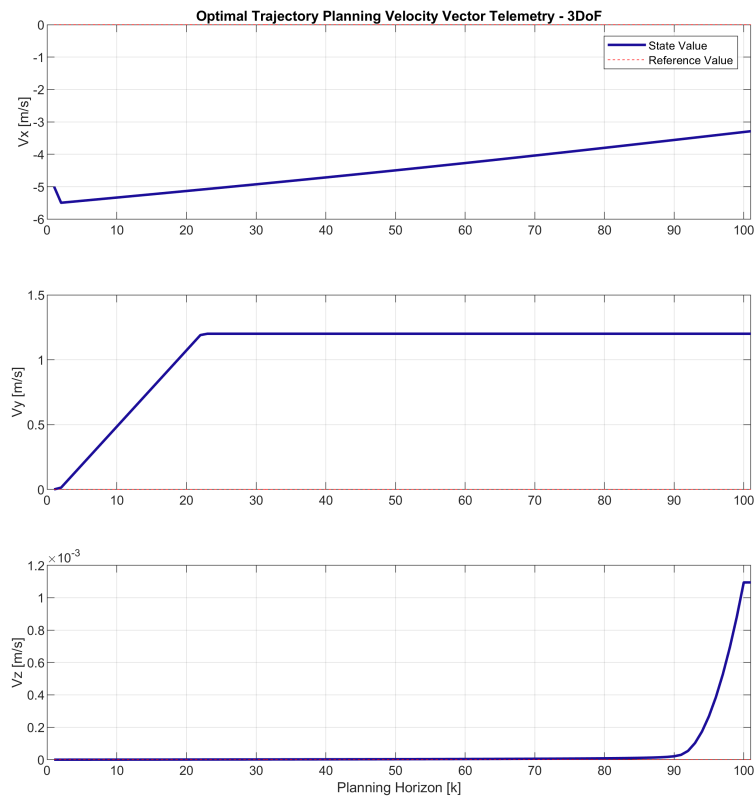


Figure 5.5: 3-DoF Rocket Model free-final-time optimal guidance for $p_0 = [30, 0, 0]$, $v_0 = [-5, 0, 0]$, $p_f = [0, 8, 0]$, $v_f = [0, 0, 0]$

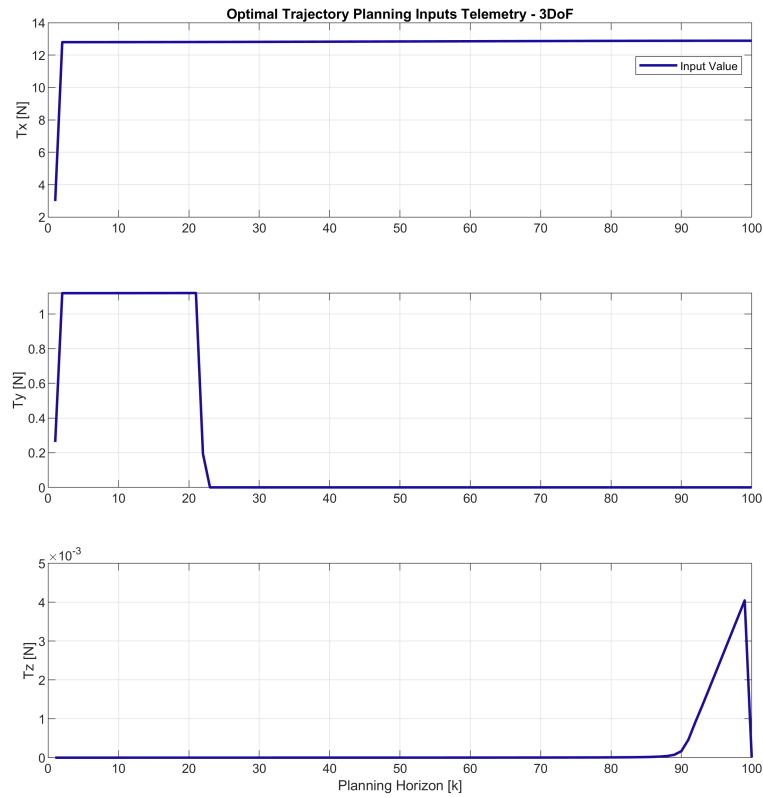
Indicator	Value
Minimal Flight Time Planned	6.6522 s
Target Point Position error	1 m
Target Point Velocity error	3.5 m/s



(a) Linear Position Telemetry



(b) Linear Velocity Vector Telemetry



(c) Thrust Vector Telemetry

Figure 5.6: First 3-DoF Rocket Model free-final-time optimal guidance telemetry

The second test of an optimal guidance for a landing maneuver requires the guidance algorithm to generate a 3D trajectory that connects the initial point $[p_x, p_y, p_z] = [50, 0, 0]$ with the target landing point $[p_x, p_y, p_z] = [0, 5, 10]$. Additionally, the vehicle is assumed to have a initial negative vertical velocity of $[v_x, v_y, v_z] = [-5, 0, 0]$ while its landing velocity should be $[v_x, v_y, v_z] = [0, 0, 0]$:

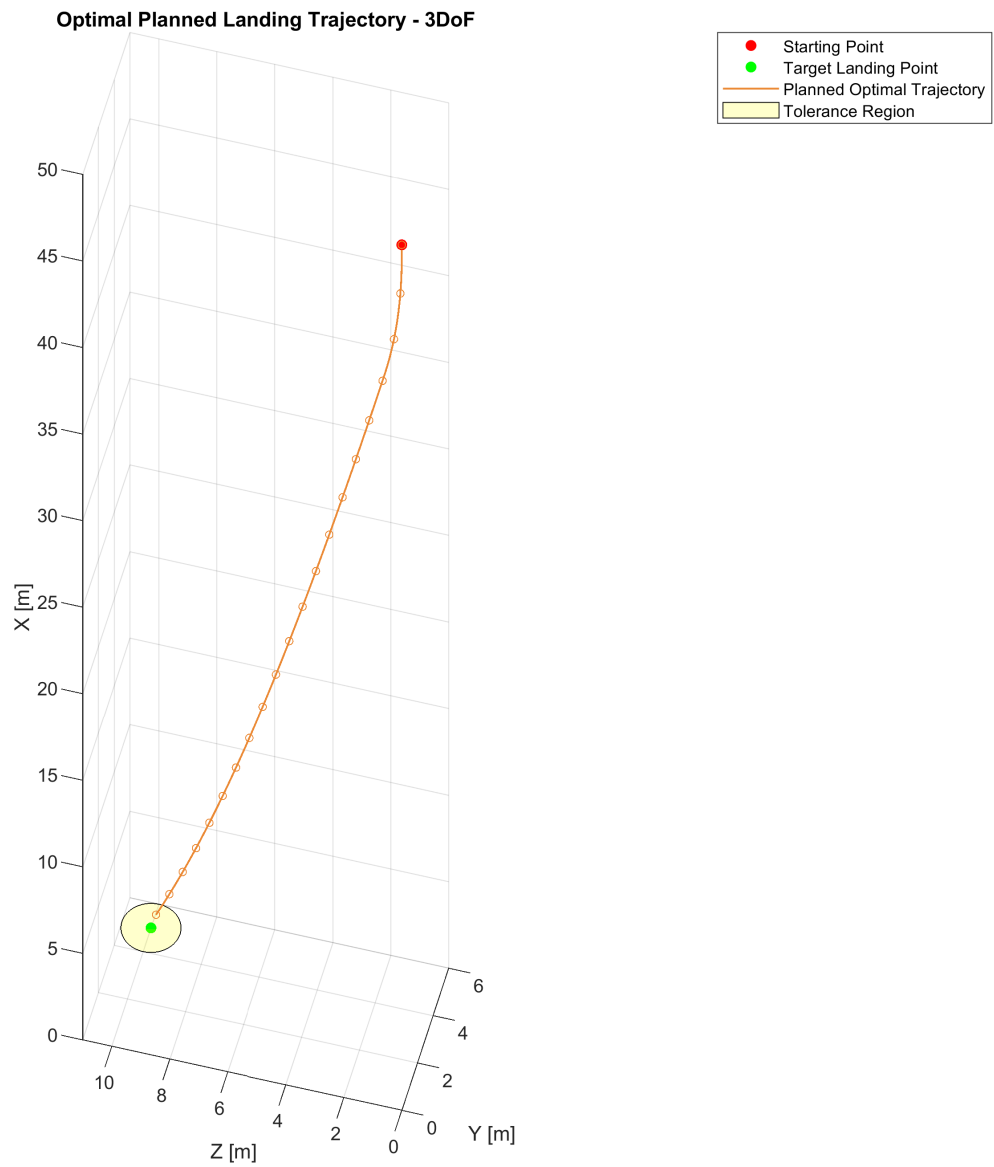
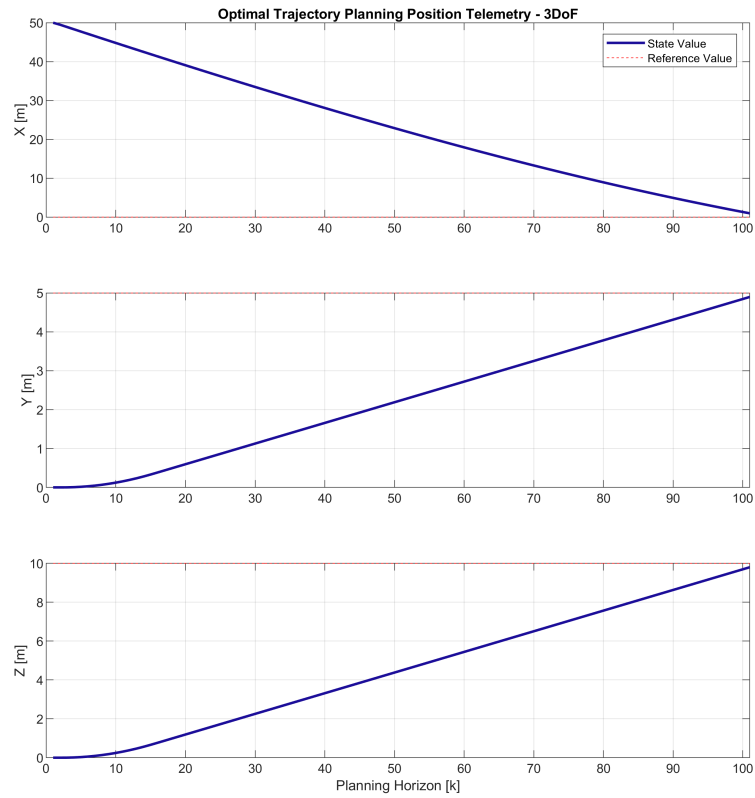
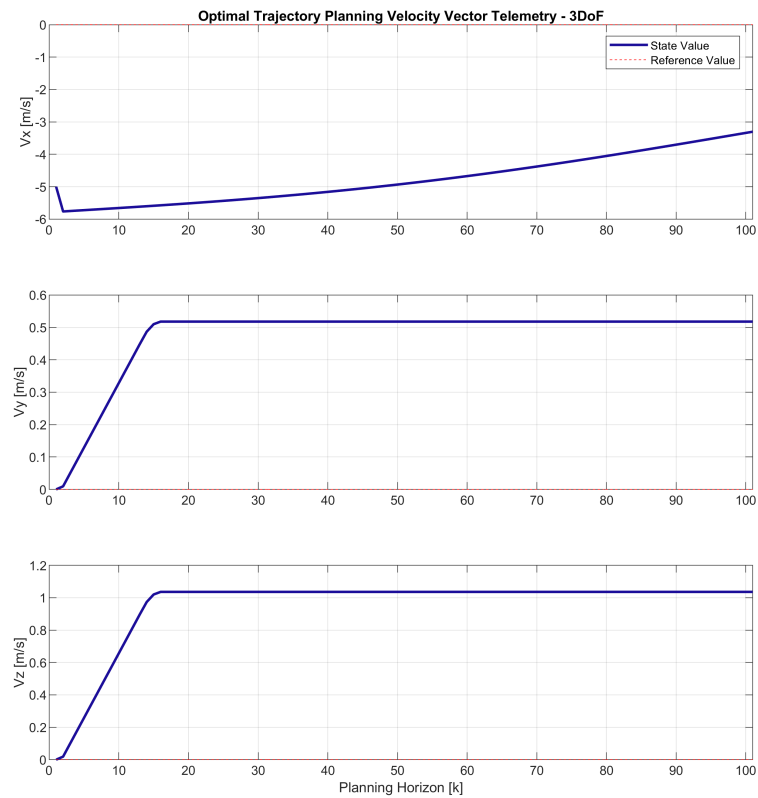


Figure 5.7: 3-DoF Rocket Model free-final-time optimal guidance for $p_0 = [50, 0, 0]$, $v_0 = [-5, 0, 0]$, $p_f = [0, 5, 10]$, $v_f = [0, 0, 0]$

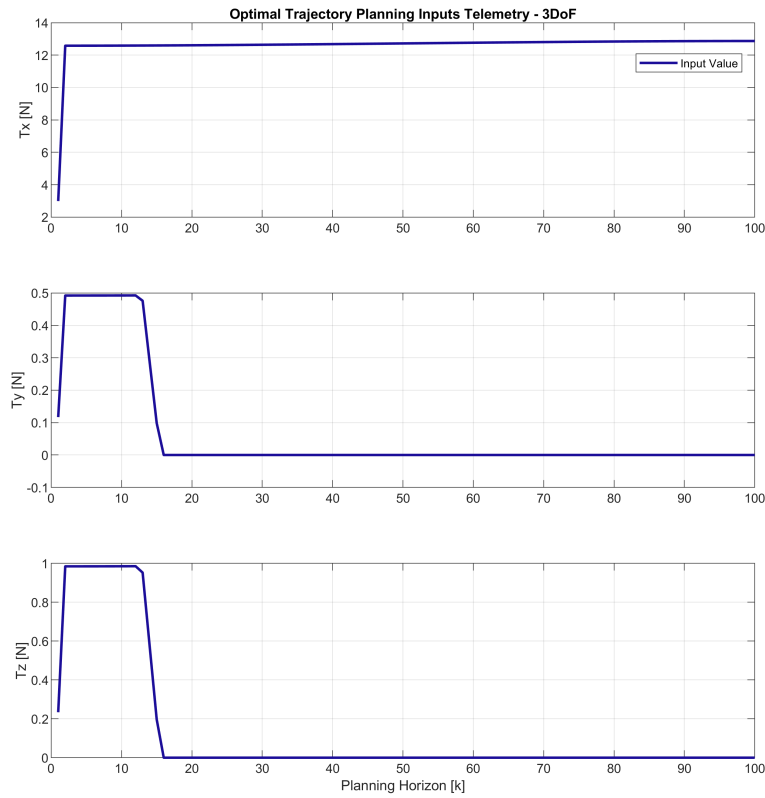
Indicator	Value
Minimal Flight Time Planned	10.2553 s
Target Point Position error	1 m
Target Point Velocity error	3.5 m/s



(a) Linear Position Telemetry



(b) Linear Velocity Vector Telemetry



(c) Thrust Vector Telemetry

Figure 5.8: Second 3-DoF Rocket Model free-final-time optimal guidance telemetry

The free-final-time optimal guidance algorithm is also tested for launch maneuvers, where the rocket is meant to execute a propelled ascent from $[p_x, p_y, p_z] = [0, 0, 0]$ with the objective of reaching a target point $[p_x, p_y, p_z] = [30, 0, 0]$ located mid air. The third test for the optimal guidance is the following:

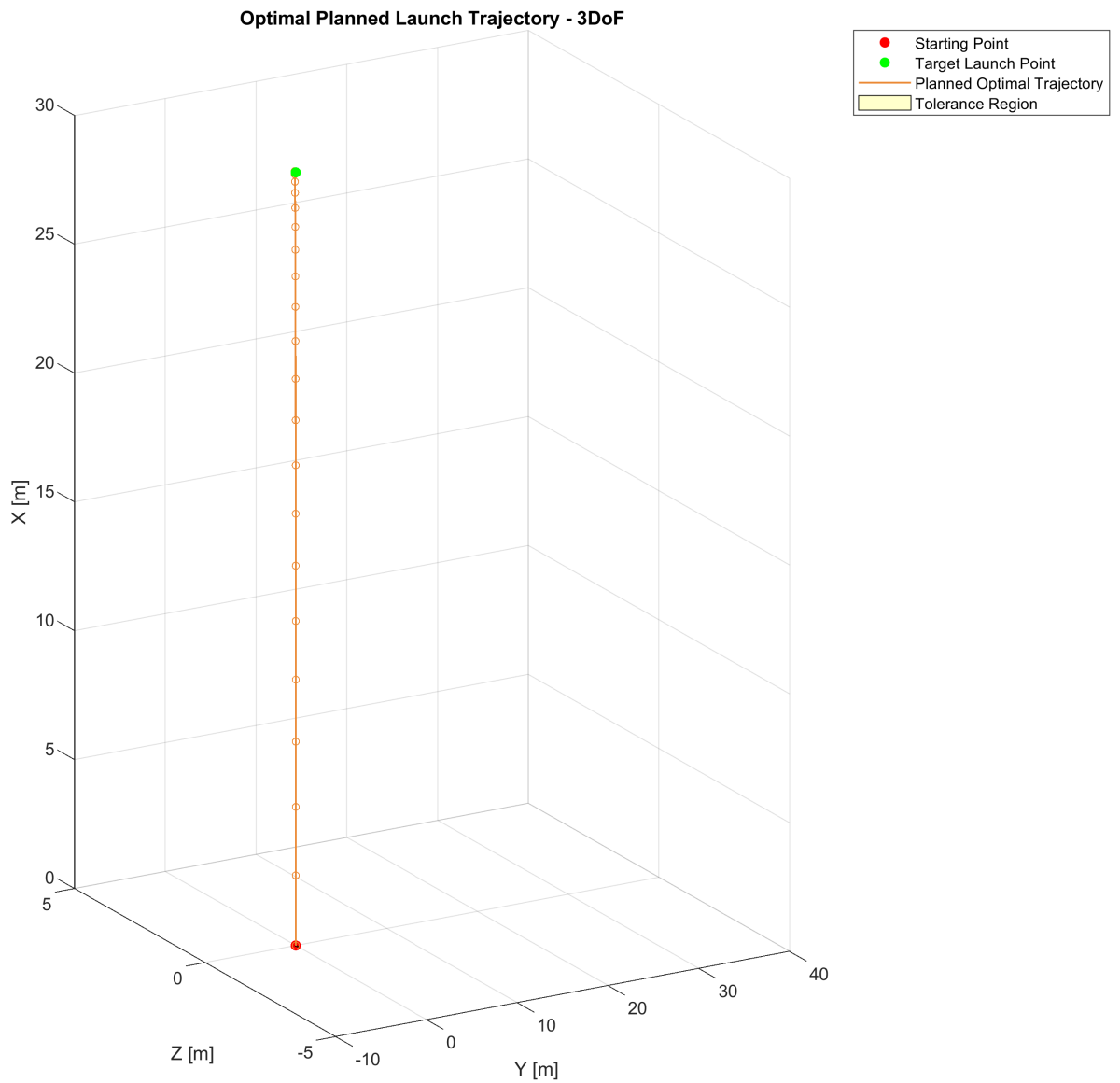
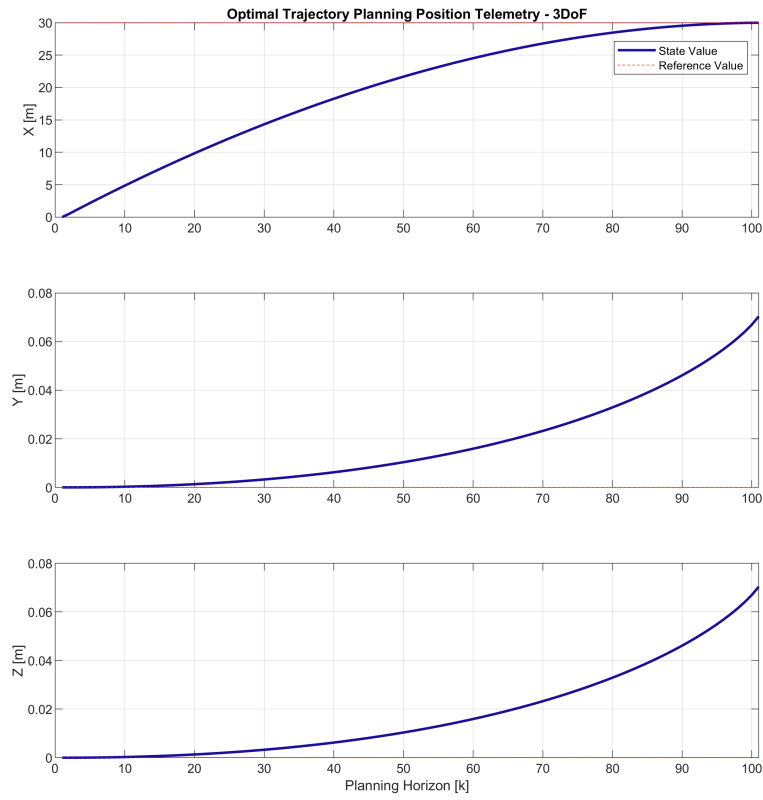
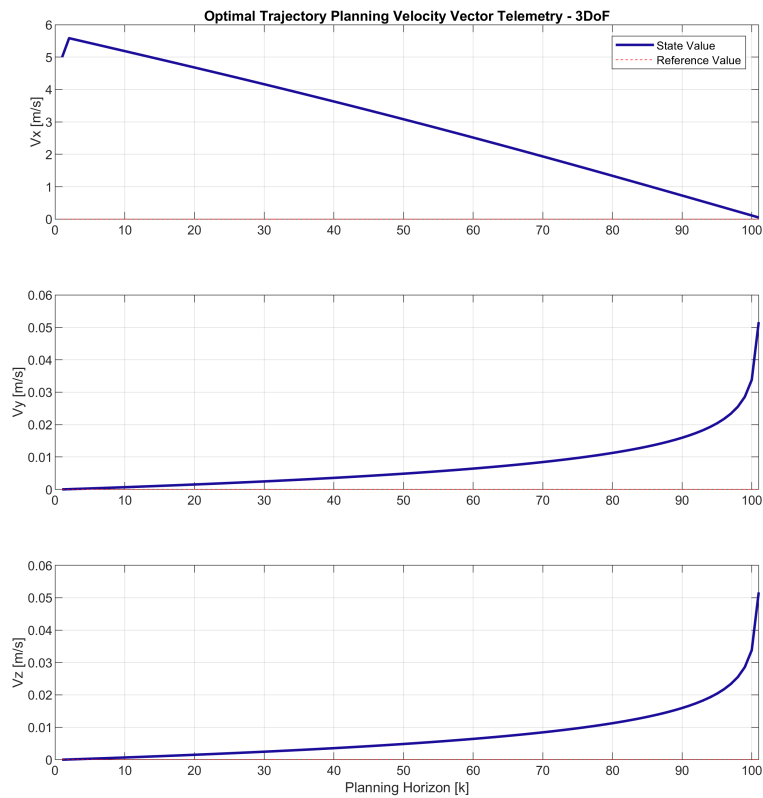


Figure 5.9: 3-DoF Rocket Model free-final-time optimal guidance for $p_0 = [0, 0, 0]$, $v_0 = [5, 0, 0]$, $p_f = [30, 0, 0]$, $v_f = [0, 0, 0]$

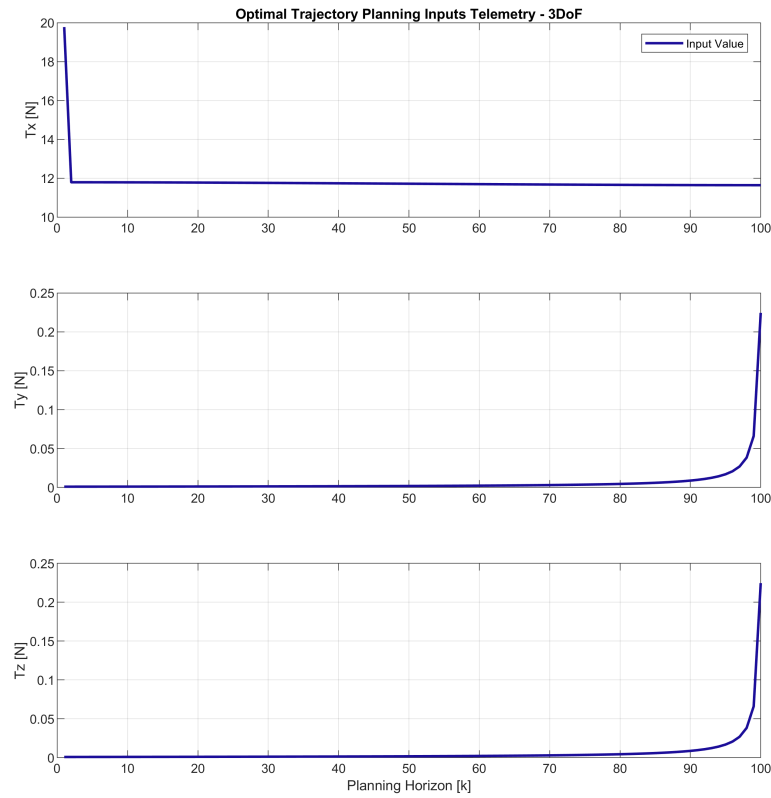
Indicator	Value
Minimal Flight Time Planned	10.0592 s
Target Point Position error	0.099487 m
Target Point Velocity error	0.088241 m/s



(a) Linear Position Telemetry



(b) Linear Velocity Vector Telemetry



(c) Thrust Vector Telemetry

Figure 5.10: Third 3-DoF Rocket Model free-final-time optimal guidance telemetry

As it can be seen, the free-final-time optimal guidance algorithm has been capable of generating landing and launch maneuvers by using the 3-DoF rocket's model and taking into account the previously described tuning for the optimal control problem parameters. In spite of this, convergence issues have been found for stricter tuning of the parameters, as well as for other more demanding boundary conditions, where the solver IPOPT has not been able to calculate a solution to the optimal control problem.

5.3 NMPC Launch/Landing Trajectory Tracking

The nonlinear model predictive position controller is tested through the tracking of some of the previously generated optimal guidance trajectories. In this case, the guidance trajectory already predefines a reference for the initial, final and intermediate waypoint's linear position and velocity of the vehicle. Additionally, initial and final conditions are specified for the rocket's attitude and angular velocity, which are assumed to be such that the rocket is completely straight and at rest at both starting point and target point.

The following problem parameters are used along all the tests in this subsection:

Parameter	Value
N	100
Hp	20
$waypoint_{p_{tol}}$	0.5 m
$waypoint_{v_{tol}}$	1 m
q_{tol}	0.1 m
ω_{tol}	0.1 m
T_{min}	3 N
T_{max}	20 N
$\dot{T}_{min}$	0 N
$\dot{T}_{max}$	10 N
δmax	5 °

In addition, and due to the fact that both the 6-DoF control-oriented and simulation-oriented models are used in these simulations, the following value for the additive disturbances is used:

$$\nu \in N(\mu, \Sigma) = N(0, 0.1)$$

The first landing trajectory tracking test is executed for [5.5](#):

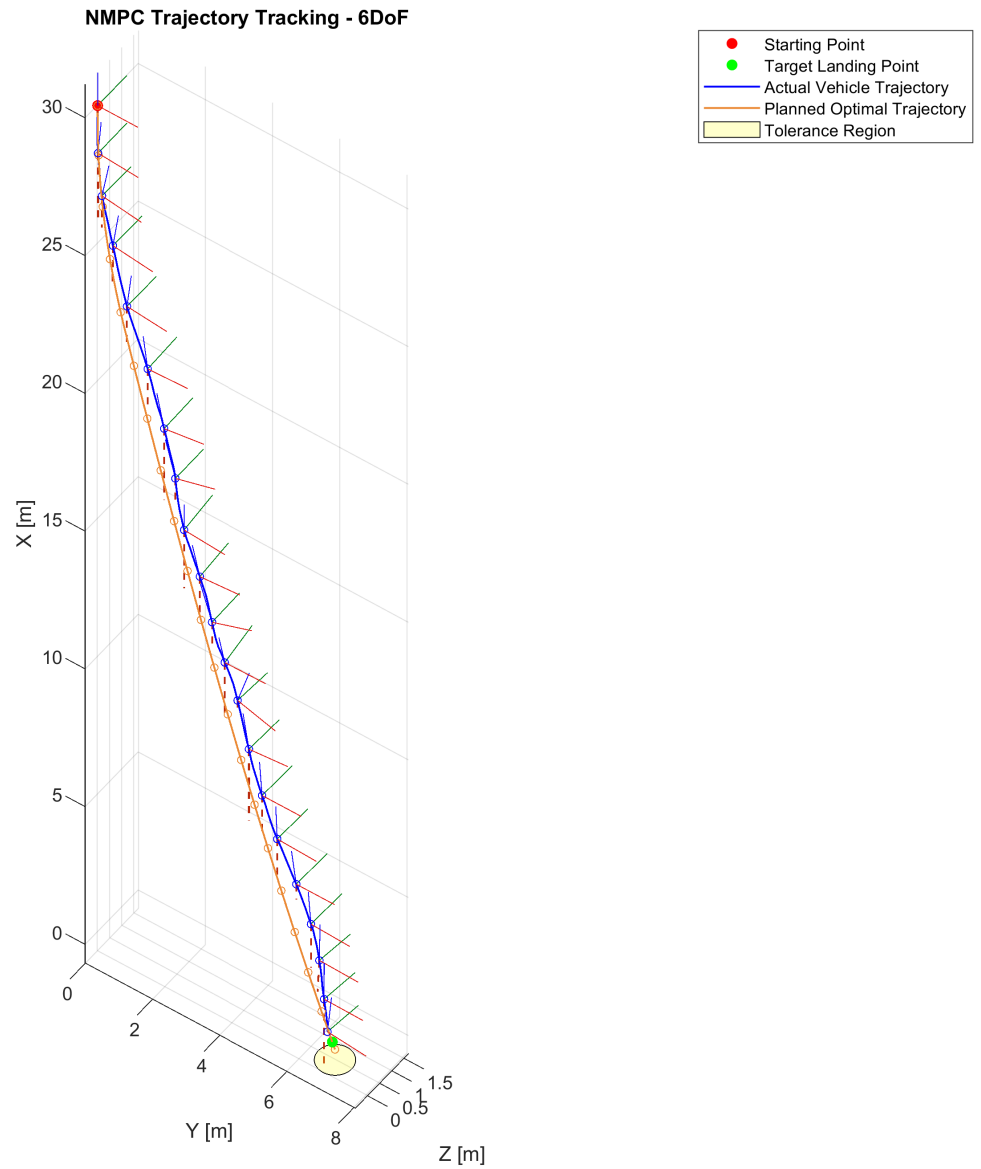
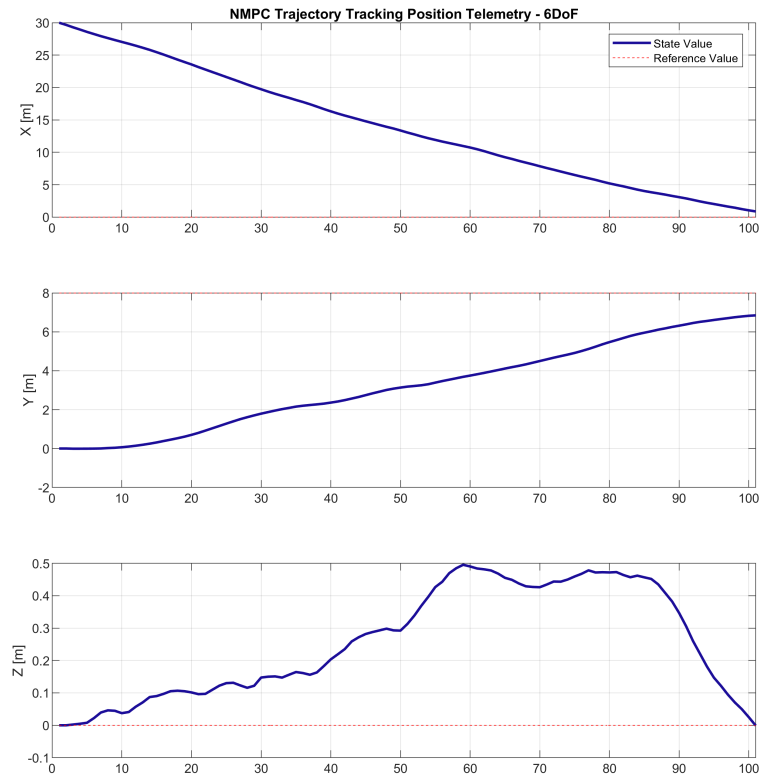
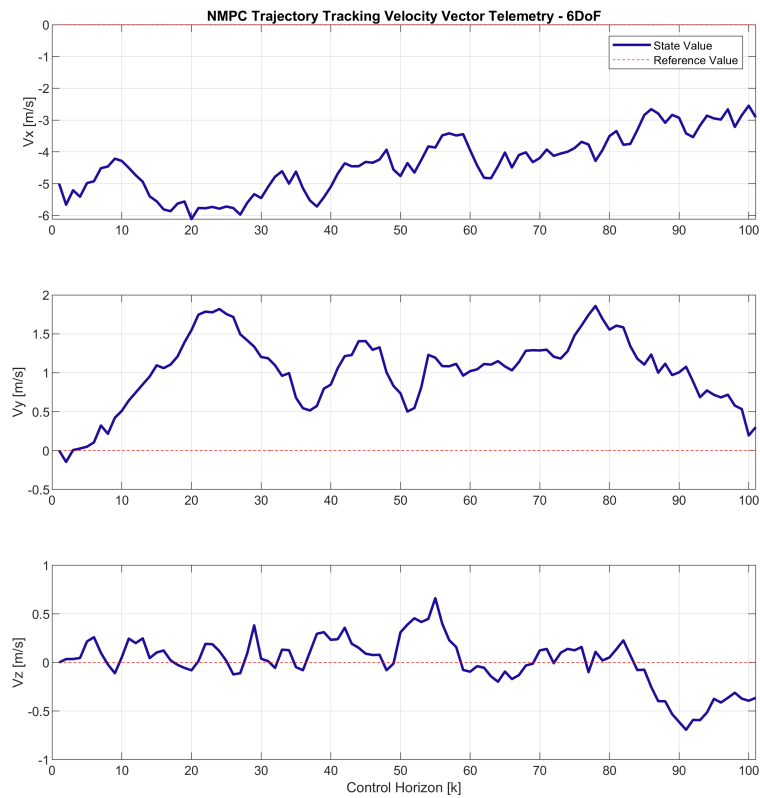


Figure 5.11: 6-DoF Rocket Model NMPC trajectory tracking for $p_0 = [30, 0, 0]$, $v_0 = [0, -5, 0]$, $q_0 = [1, 0, 0, 0]$, $\omega_0 = [0, 0, 0]$, $p_f = [0, 8, 0]$, $v_f = [0, 0, 0]$, $q_f = [1, 0, 0, 0]$, $\omega_f = [0, 0, 0]$

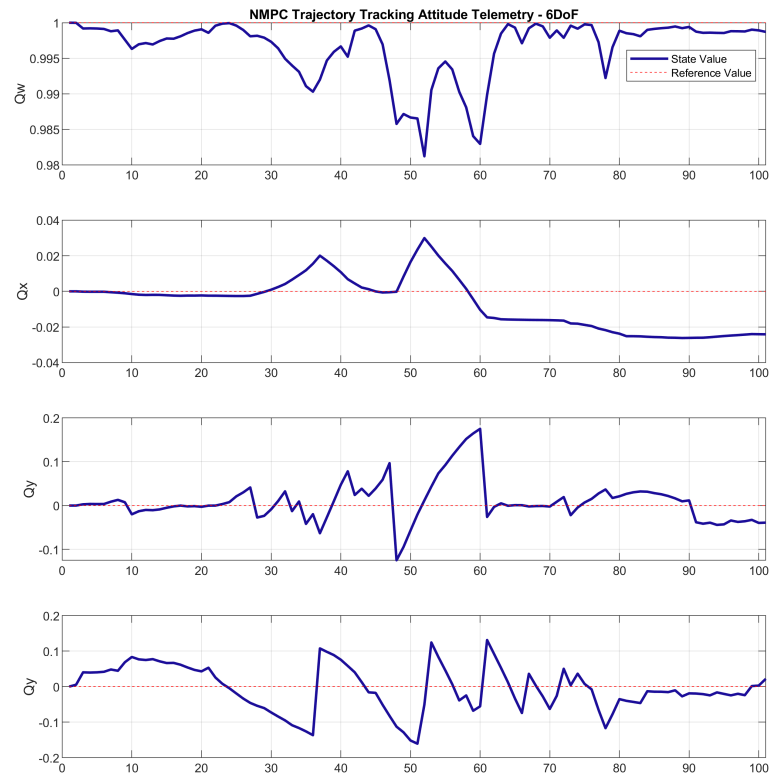
Note: The evolution of the rocket's attitude is represented by the different vehicle frames that appear along the flight path. Initially, the rocket is vertically standing and pointing up, but rotates along the flight path as a result of the TVC actuator.



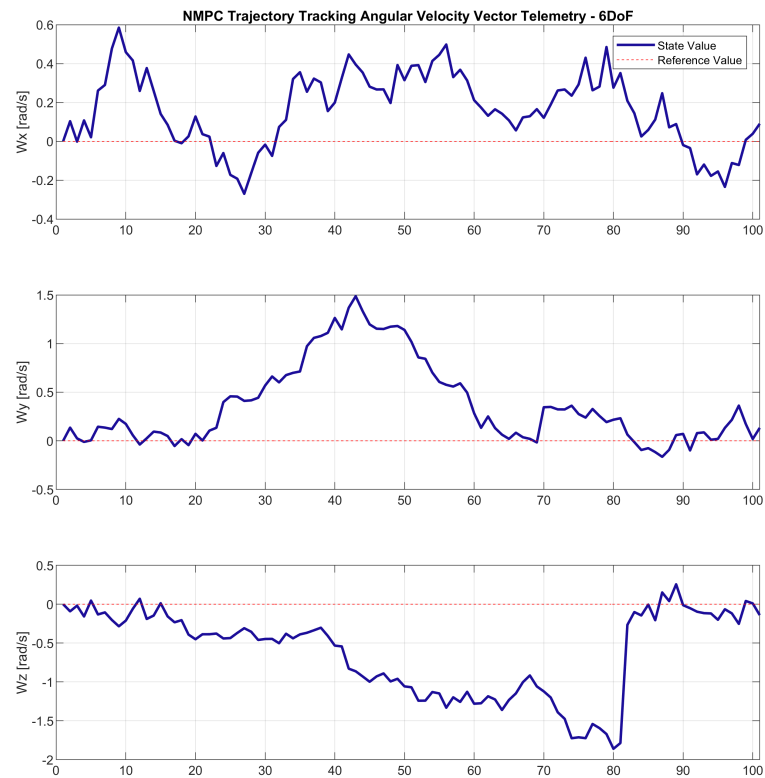
(a) Linear Position Telemetry



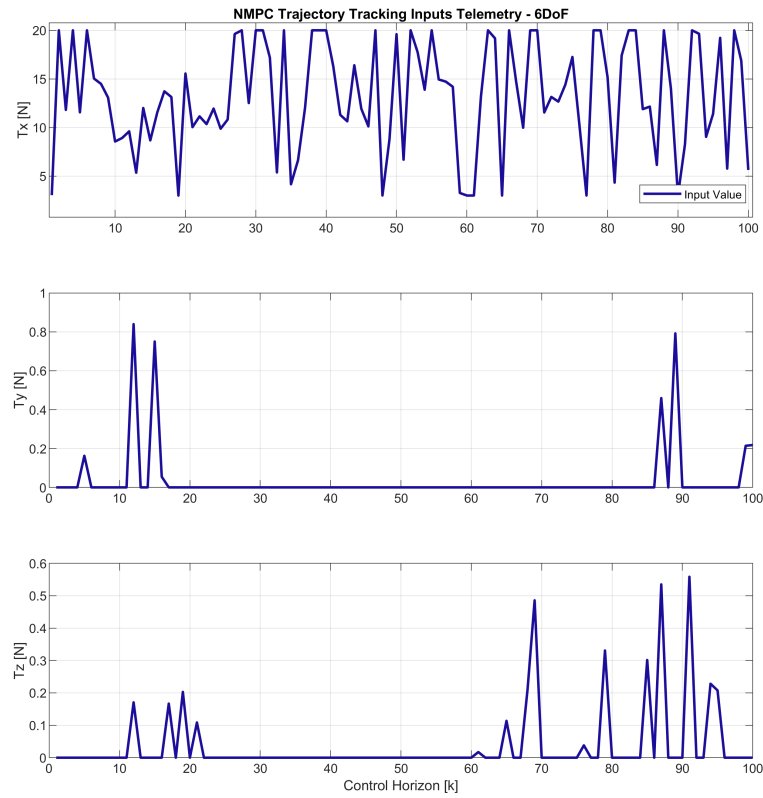
(b) Linear Velocity Vector Telemetry



(c) Attitude Quaternion Telemetry



(d) Angular Velocity Vector Telemetry



(e) Thrust Vector Telemetry

Figure 5.12: 6-DoF Rocket Model Closed Loop First Trajectory Tacking Telemetry

The following error statistics are obtained:

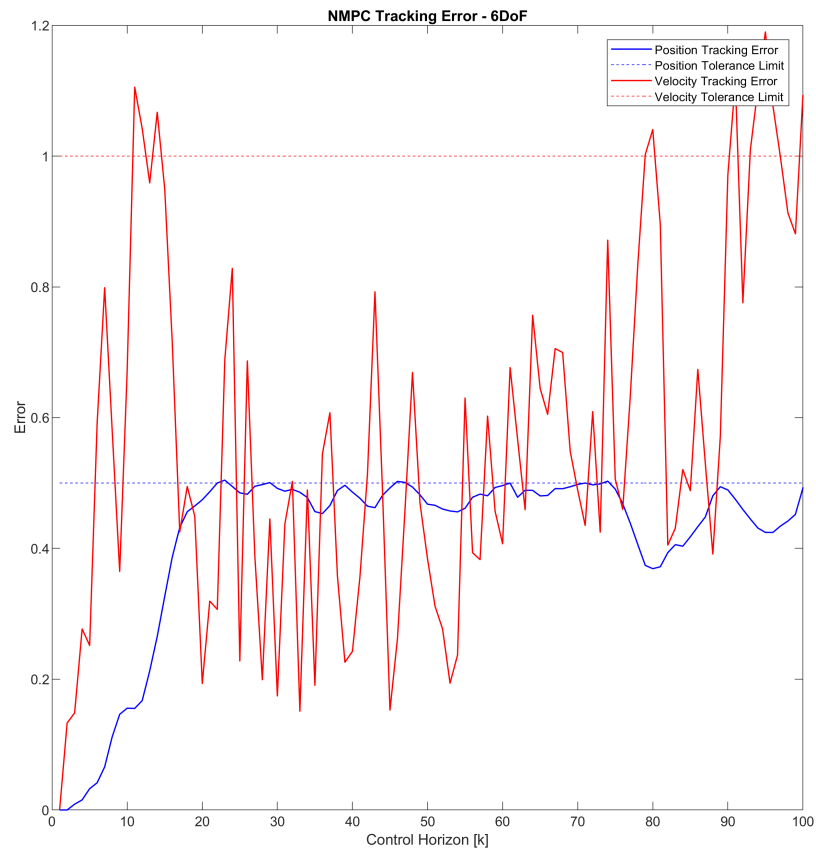


Figure 5.13: 6-DoF Rocket Model NMPC first trajectory tracking error

Indicator	Value
Average position error along trajectory	0.40409 m
Average velocity error along trajectory	0.60946 m/s
Landing Position error	0.54556 m
Landing Velocity error	1.0419 m/s

The second landing trajectory tracking test is the following [5.5](#):

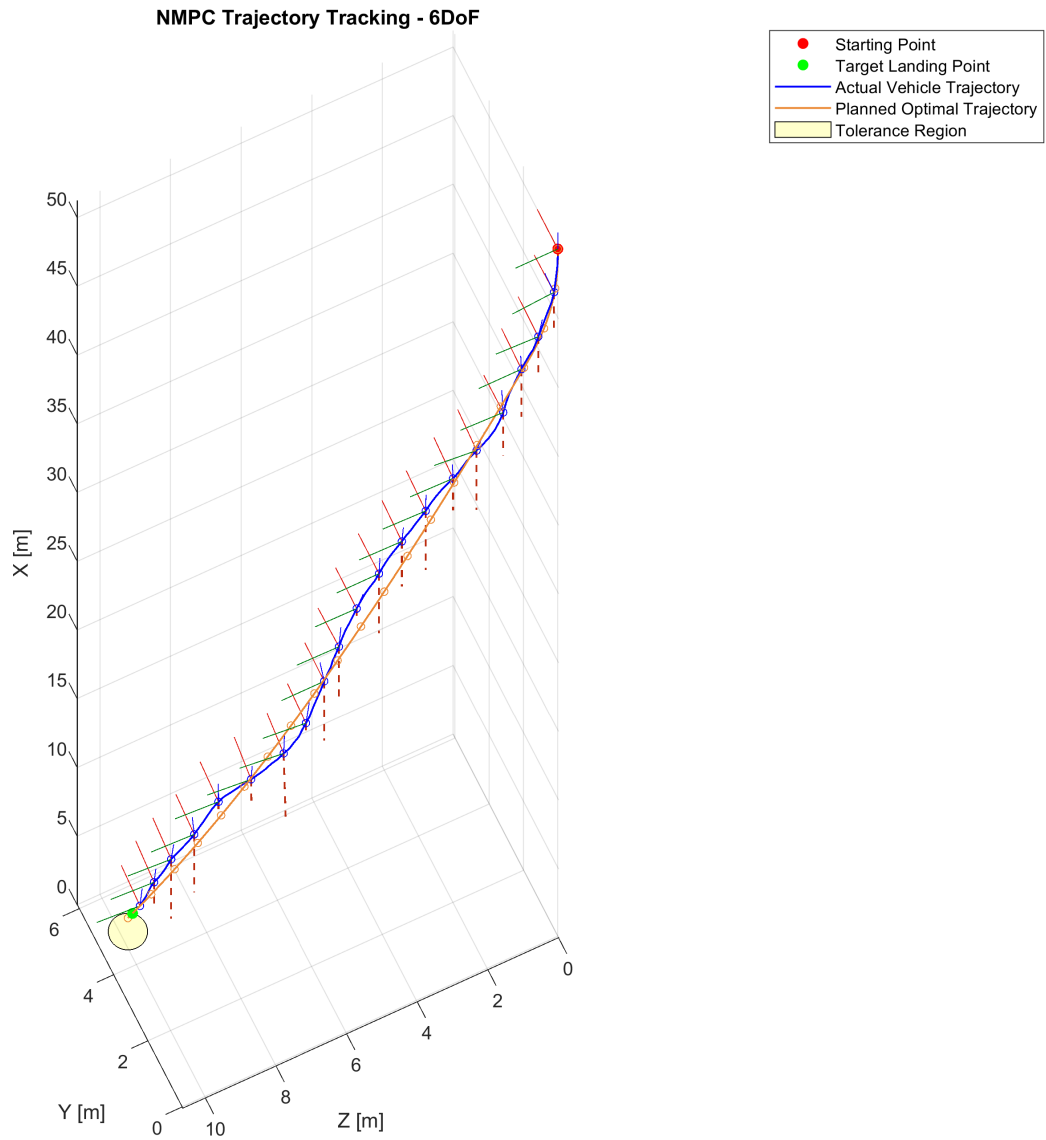
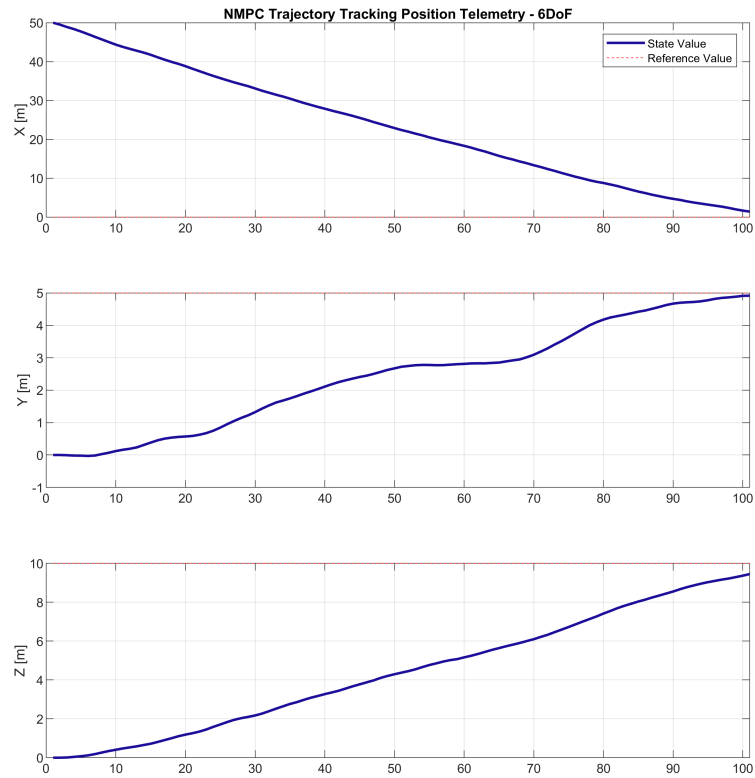
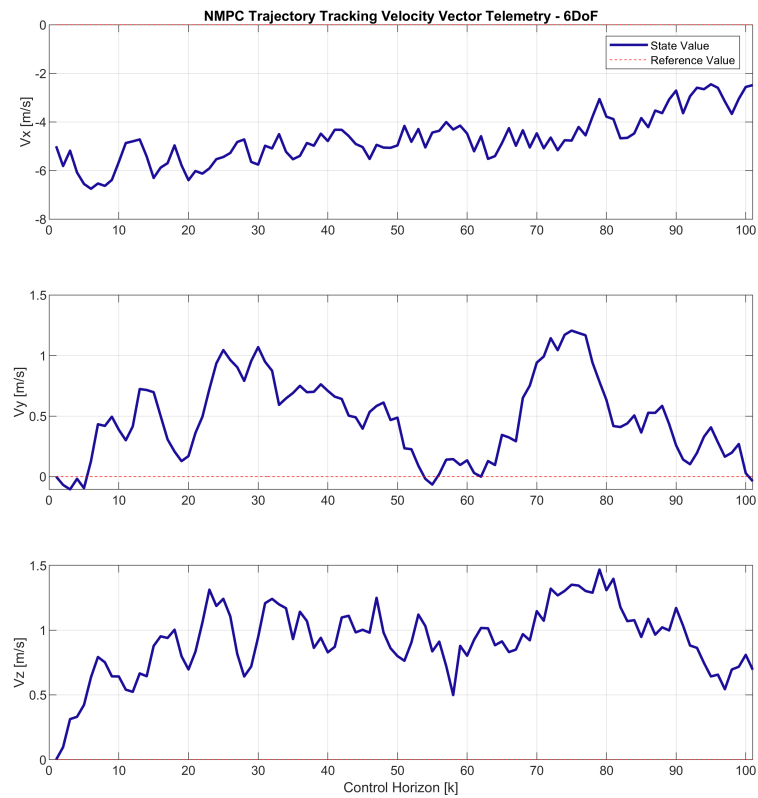


Figure 5.14: 6-DoF Rocket Model NMPC trajectory tracking for $p_0 = [50, 0, 0]$, $v_0 = [0, -5, 0]$, $q_0 = [1, 0, 0, 0, 0]$, $\omega_0 = [0, 0, 0]$, $p_f = [0, 5, 10]$, $v_f = [0, 0, 0]$, $q_f = [1, 0, 0, 0, 0]$, $\omega_f = [0, 0, 0]$

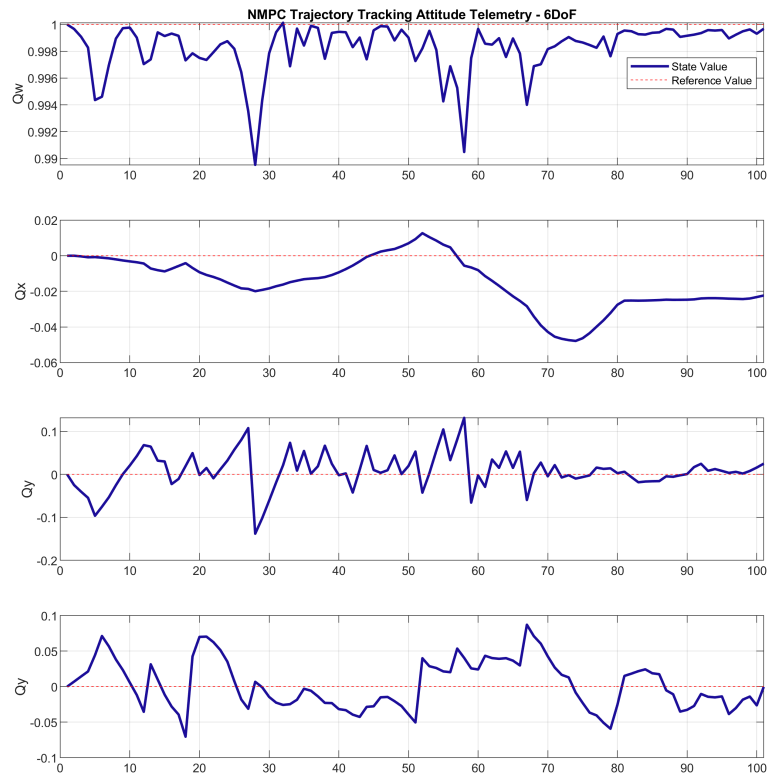
Note: The evolution of the rocket's attitude is represented by the different vehicle frames that appear along the flight path. Initially, the rocket is vertically standing and pointing up, but rotates along the flight path as a result of the TVC actuator.



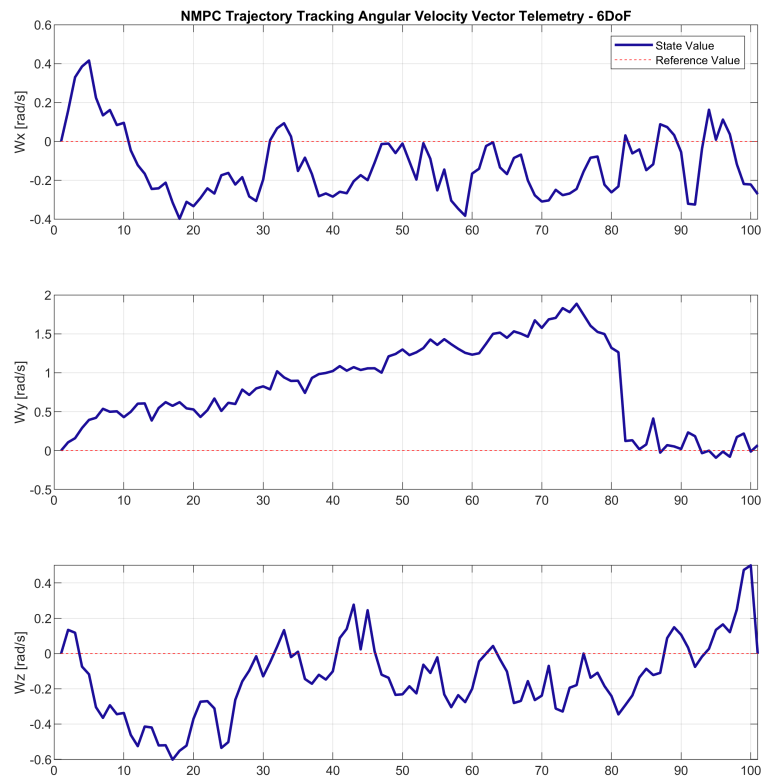
(a) Linear Position Telemetry



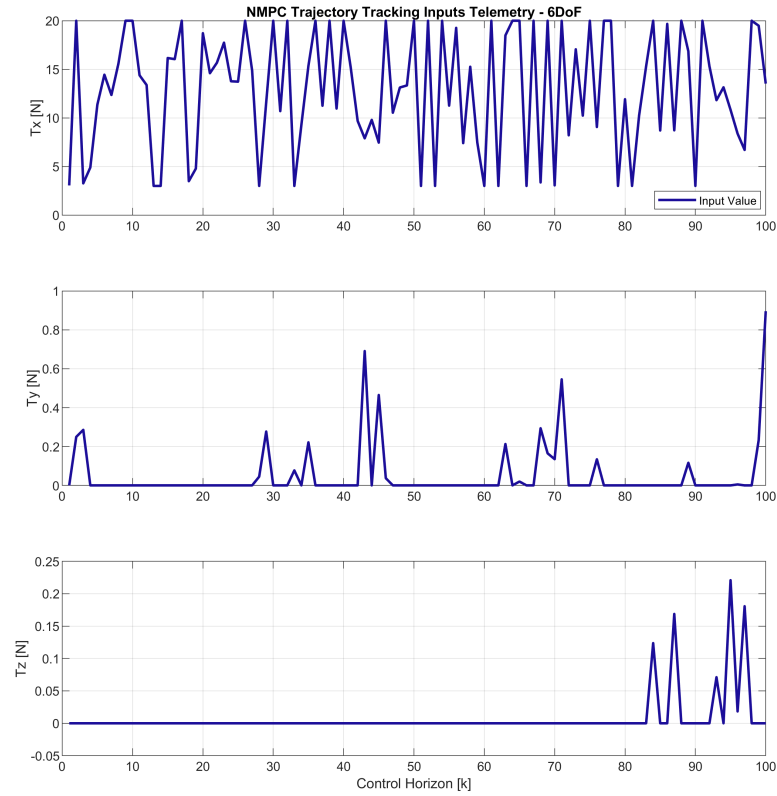
(b) Linear Velocity Vector Telemetry



(c) Attitude Quaternion Telemetry



(d) Angular Velocity Vector Telemetry



(e) Thrust Vector Telemetry

Figure 5.15: 6-DoF Rocket Model Closed Loop Second Trajectory Tacking Telemetry

The following error statistics are obtained:

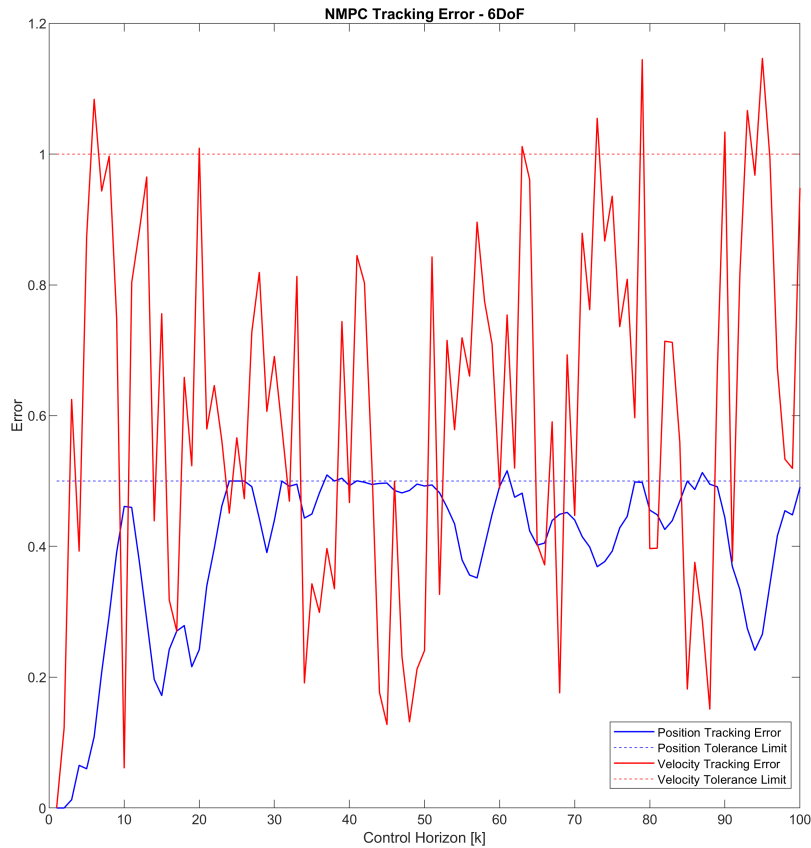


Figure 5.16: 6-DoF Rocket Model NMPC first trajectory tracking error

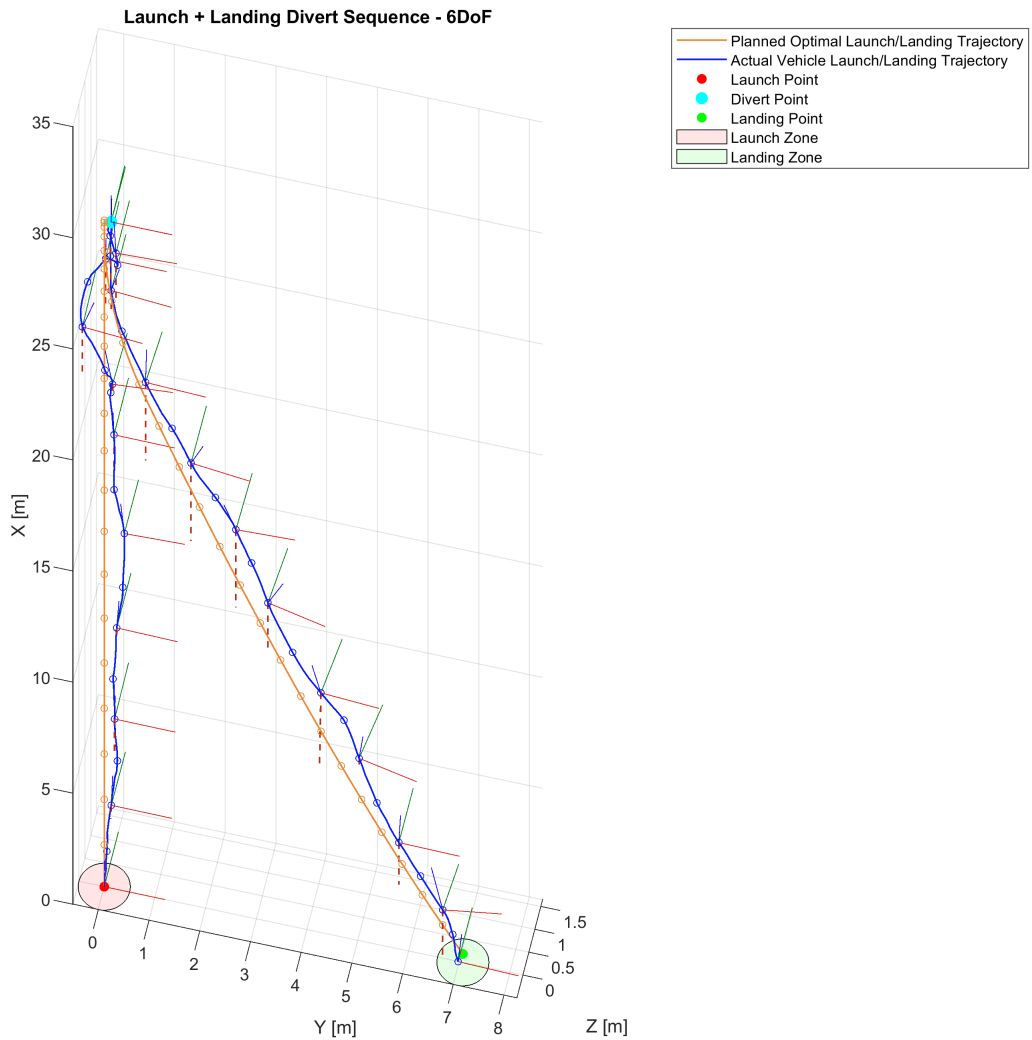
Indicator	Value
Average position error along trajectory	0.40172 m
Average velocity error along trajectory	0.61394 m/s
Landing Position error	0.55967 m
Landing Velocity error	1.0455 m/s

As it can be seen, the nonlinear model predictive controller algorithm has been capable of successfully tracking reference landing maneuvers by using the 6-DoF rocket's model under disturbance conditions. During these simulations, the optimization problem parameter tuning was found to be a key feature of the algorithm affecting its capability to successfully control the motion of the rocket. In spite of this, convergence issues have been found for stricter tuning of the parameters, as well as for other more demanding boundary conditions, where the solver IPOPT has not been able to calculate a solution to the optimal control problem. Lastly and as observed in figures 5.13, 5.16, the position and velocity tracking tolerances exceed their respective limits in some specific instants, which may be caused by the effect of the additive disturbance in the simulation model, given that the NMPC does not use any disturbance rejection method.

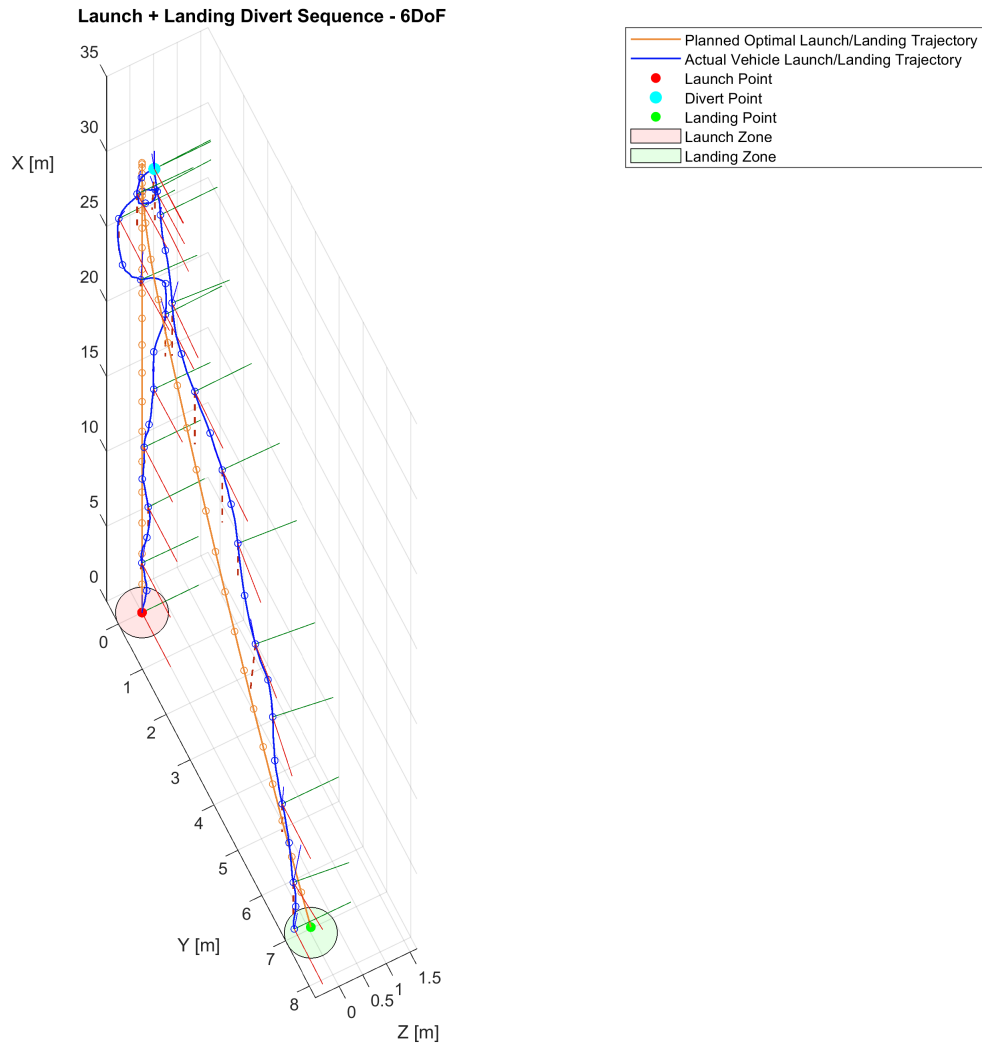
5.4 On-Flight Divert Simulation

On-flight divert maneuvers represent one of the most challenging situations within the rocket landing GNC problem. This type of maneuvers require in the rocket to do an initial take-off maneuver that is interrupted by a mid-flight command that requests the rocket to abort its ascent to perform a landing maneuver on a nearby target location. This is a standard test to validate the performance of landing vehicles, as it can be seen in [7, 8, 29].

With the final objective of performing an integrated rocket launch and land test, a simulation in which the vehicle is requested to do a divert sequence has been executed using the previously described setups for the guidance and control algorithms, producing the following results:



(a) 6-DoF Divert Sequence Trajectory, Lateral View



(b) 6-DoF Divert Sequence Trajectory, Top View

Figure 5.17: 6-DoF Divert Sequence Trajectory for $p_0 = [0, 0, 0]$, $v_0 = [0, 0, 0]$, $q_0 = [1, 0, 0, 0]$, $\omega_0 = [0, 0, 0]$, $p_{divert} = [30, 0, 0]$, $v_{divert} = [2, 0, 0]$, $q_{divert} = [1, 0, 0, 0]$, $\omega_{divert} = [0, 0, 0]$, $p_f = [0, 8, 0]$, $v_f = [0, 0, 0]$, $q_f = [1, 0, 0, 0]$, $\omega_f = [0, 0, 0]$

5.5 Simulation Insights

The key insights that have been concluded from all the simulations performed in this section of the thesis are:

- As depicted in the vehicle's models open loop simulations, the behavior of the 3-DoF model differs from the 6-DoF model one, as the latter shows a closer-to-reality thrust vector controlled characteristic rocket motion. This happens due to the absence of rotation and attitude dynamics in the oversimplified 3-DoF vehicle's model.
- Despite the previous statement, the 3-DoF model has proved to be useful when used as a guidance model, as its translational dynamics are sufficiently precise for generating 3D trajectories subject to a given set of flight path constraints.
- The 6-DoF, more complex vehicle model is required when performing position control in order to ensure a realistic calculation of thrust vector inputs.
- The free-final-time optimal guidance algorithm has proved to be effective, but is really sensitive to the tuning of the parameters involved in the optimization problem, as IPOPT solver convergence problems have been found for some demanding scenarios. In these cases, the optimization problem cannot be solved, hence no feasible trajectory with minimal final time can be generated. In order to fix this issue, the approach followed in this work consists in relaxing the constraints of the problem by seeking a trade-off between feasibility and optimality.
- In the case of the nonlinear model predictive position controller, a similar situation has been found. The algorithm is really sensitive to the tuning of the optimization parameters and shows solver convergence problems for some demanding scenarios. A special mention is done for the prediction horizon parameter, which causes a bad controller performance for very low values ($Hp \leq 10$), but shows adequate results for not-so-high values ($Hp \geq 20$). However, it has been observed that the solution found by the NMPC algorithm (even when convergence fails) is usually good enough and can be applied to the system, showing its capability to successfully control the rocket. However, convergence issues may be also related to the tolerance surpassing effect depicted in [5.13, 5.16](#).
- Both previously mentioned problems are a consequence of the nonlinear and nonconvex formulations of the guidance and control problems, which pose serious challenges in terms of ensuring solution convergence, feasibility and stability when solving optimal control problems. Due to this fact, a convexification process should be applied to both problems in order to improve their performance, as discussed in [\[5, 6, 30\]](#).
- The implementation done in this work using Matlab and CasADi shows a heavy compu-

tational burden that requires rather high computational power and is therefore valid for the design and simulation phases, but not for real-life on-board applications, where other specific real-time tools should be used.

Chapter 6

Effects on Economy, Society and Environment

This section presents an overview of the potential impacts of reusable rockets on the economy, society, and environment. It is divided into two parts: socioeconomic effects and environmental effects.

6.1 Socioeconomic impact

The introduction of rocket reusability technology within the aerospace industry sector represents a game changer situation that has some relevant effects in society and economics.

In relation to economics, the main impact of reusability of space rockets is a drastic cost reduction in space launch operations. An example of this is discussed in [31], where the SpaceX Falcon Heavy reusable rocket is concluded to reduce launching costs by a 65% when compared with other similar category non reusable rockets. Furthermore, the reduction of launching costs triggers a space launch market expansion, as the demand of this kind of services increases due to the fact that the access to space becomes cheaper, making it more accessible for startup, research centers, etc. Another consequence of the previously mentioned impacts is the creation of many new jobs positions within aerospace, manufacturing, software development, logistic and launcher maintenance operations.

In relation to the social impact, the push for reusable rockets generate huge technological advancements that generates potential collateral improvement effects in other industries, as well as in the commercial services that are offered to the population. An example of this is the SpaceX Starlink project that uses the reusable launcher Falcon 9, which facilitate the deployment of large satellite constellations, providing global internet coverage, especially in remote and undeserved regions. In addition, the success of reusable rockets serves as a source of inspiration for future

generations, increasing people's interest in science, technology, engineering, and mathematics (STEM) fields, also providing educational opportunities, as more people can engage with space exploration and its possibilities.

6.2 Environmental impact

The environmental impact of reusable rockets can be both positive and negative. In order to give a global view, some advantages and disadvantages are discussed hereafter:

Given the fact that most reusable rockets can resist several launch/land operations, as for instance the Blue Origin New Shepard that has successfully completed 26 missions, reusability technology generates a good environmental impact by reducing consumption of raw rocket manufacturing materials. Another important point where reusability helps the environment is due to the reduction in space debris, as rocket booster first stages return back to earth in a controlled manner. Finally, the design and implementation of new reusable rocket presents a nice opportunity for developing new propulsion fuels that are more environmentally friendly in comparison with some classical jet fuels.

In spite of the previously mentioned, reusable and non reusable rockets are still contaminant and produce atmospheric pollution, as discussed in [32]. The recent rise in the number of space rocket launch operation may suppose a threat to the environment, and therefore, should be carefully monitored by the space industry authorities in order to ensure its sustainability in the long term.

Chapter 7

Project Budget

Given the fact that this work has been carried out in a simulation setup, the expenses related to it are few, as no building cost exist. In order to determine the total cost of the project, the following concepts are considered:

- Matlab license cost: A standard license for Matlab full access costs 900 €.
- Computing platform cost : A powerful workstation PC used to run the guidance and control algorithms simulations. In this work, the MSI Trident 3 model was used, which cost around 1300 €.
- Working hours cost: Given the fact that the Master Thesis is weighted as 12 ECTS, a total effective time of 300 hours of work is done. By assuming an average engineering wage per hour cost of 14.75 €, the total working hour cost is 4425 €.

A hypothetical total cost of the project is calculated as follows:

$$\text{Total Cost} = \text{Matlab License} + \text{Computing Platform} + \text{Working Hours}$$

$$\text{Total Cost} = 900 + 1300 + 4425 = 6625\text{€}$$

Furthermore, in the case this work was to be tested in a real life experimental setup, it will be required to built a rocket-like small-scale VTOL platform similar to the one described in [1], as the physical parameters of that vehicle were the ones used to theoretically simulate the rocket of this work. This situation would add around 907 € to the total cost.

Chapter 8

Concluding Remarks

8.1 Conclusions

This master's thesis has been dedicated to studying some of the mathematical modelling and control methods that are used nowadays to approach the autonomous vertical precision landing problem of space rockets from the control engineering perspective. All the work has been done in a theoretical manner and has been tested using computer simulations.

Initially, the physical parameters of the EmborockETH small-scale VTVL lander [1], a physically built vehicle, were used as a reference for the modelling of the theoretical rocket used in this work. This vehicle is controlled through thrust vectoring by swiveling the main thruster and therefore, inducing forces and torques around the center of mass of the vehicle. Two different dynamic models were derived in order to mathematically represent the motion of the vehicle: a three degrees of freedom oversimplified model, in which only the translational dynamics of the rocket motion are considered using Newton's second law; and a more complex six degrees of freedom model, which takes into account the translational, rotational and attitude dynamics of the vehicle through the use of rigid body physics and quaternion theory. The aerodynamic effects are neglected in the modelling process.

Using the previously derived dynamic models, two control algorithms were formulated: a free-final-time optimal guidance system and a nonlinear model predictive position control system. The objective of the guidance system was to generate trajectories that connected an initial and target point in the 3D space by ensuring minimum flight time, minimum control effort and the verification of a set of flight path constraints. The guidance algorithm was formulated as a model-based optimal control problem using the three degrees of freedom dynamics and was solved using an open-loop approach. Then, the nonlinear model predictive position controller used the reference trajectories generated by the guidance system to calculate a sequence of thrust vector inputs that allow the vehicle to track launch or landing maneuvers with minimal error.

The position control algorithm was formulated as an optimal control problem that was solved using an MPC scheme, and used two different vehicle's model: the six degrees of freedom dynamics as predictive model, and an augmented six degrees of freedom disturbance model to simulate a more realistic system in the closed loop.

Finally, both algorithms were implemented using Matlab as programming interface and CasADi with IPOPT solver as nonlinear optimization modelling tool. Using the developed code, the calculation of guidance trajectories and the execution of them through the position controller were successfully simulated in diverse launch, landing and divert sequence scenarios. A tuning process of the specific parameters of both the guidance and control algorithms was required in order to make them perform as expected.

After the completion of the previously described milestones, the main conclusions that can be drawn are the following:

- The mathematical complexity required to describe the mechanics of a thrust vector controlled rocket in 3D increases a lot with respect to the analogous 2D case.
- The six degrees of freedom dynamic model of a thrust vector controlled rocket in 3D is highly nonlinear, and this presents a huge challenge when attempting to use other than nonlinear control techniques to regulate its motion.
- Although unitary quaternions represent a more sophisticated and less intuitive method for describing the 3D orientation of the rocket when compared to Euler angles, they add robustness and mathematical simplicity to the vehicle's mathematical models.
- The manipulation of differential equation's time scale is a key idea when dealing with minimum time optimal control problems. In this manner, an initially implicit parameter of the model as the final time can become explicit with the objective of using it as an optimization variable.
- Splitting the rocket motion control problem into two subproblems (guidance and position control) is a key idea for achieving successful results. In a separated way, both problems are easier to approach than in combination.
- Model-based optimal control has proved to be a powerful tool when tackling the calculation of rocket landing trajectories, given the necessity of introducing vehicle dynamics, flight path constraints and optimal criteria.
- Model predictive control has proved to be an effective method of performing multivariable, nonlinear and disturbance resistant rocket's position control.
- Matlab with CasADi setup has demonstrated to be a practical way of implementing rocket

guidance, navigation and control simulations, even though the implementation presented in this work was found to be heavy in terms of computational burden.

- When implementing optimal control techniques, the underlying solver becomes a key component of the solution that can make a notable difference in terms of feasibility and convergence. In order to obtain good results, the decision of which type of solver to use needs to be accurately reasoned.
- In relation to the previous fact, the tuning of the optimization parameters has been concluded to be really sensitive. The tuning of some of the parameters appearing in this work, such as error tolerances, prediction horizon and weighting factors, is critical and can seriously affect solver performance.

8.2 Future Work

Some of the future work ideas that emerged during the realization of this master are presented hereafter:

- To make the six degrees of freedom rocket dynamic model even more physically realistic by adding the aerodynamic effects of the atmosphere, which could be used to add new actuators and therefore increase the control authority.
- To formulate a linear parameter-varying representation of the nonlinear six degrees of freedom dynamic model, which would allow the use of linear and robust control theory results.
- To implement a more sophisticated guidance algorithm using the six degrees of freedom dynamics in order to compare it to the existent one and conclude if any remarkable improvements are found.
- To refine the optimization parameters and cost functions of both the guidance and position control algorithms in order to improve its performance.
- To improve the augmented model by making it account for a wide range of real-life disturbances and noises.
- To implement a Kalman filter in order to perform full state estimation from partial state measurement of the flight data of the rocket.
- To implement a disturbance observer with the objective of providing the position controller with disturbance rejection capabilities.
- To reformulate both the guidance and position control algorithms using the results de-

scribed in [5], in order to increase the theoretical guarantees of the implementation done in this work.

- To implement new rocket landing simulations using more powerful state of the art commercial optimization solvers as Embotech ForcesPro [28], in order to test if the guidance and control algorithm can achieve real-time performance capabilities.
- To implement all this work in an experimental setup and to test it using a real-life rocket vehicle.

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